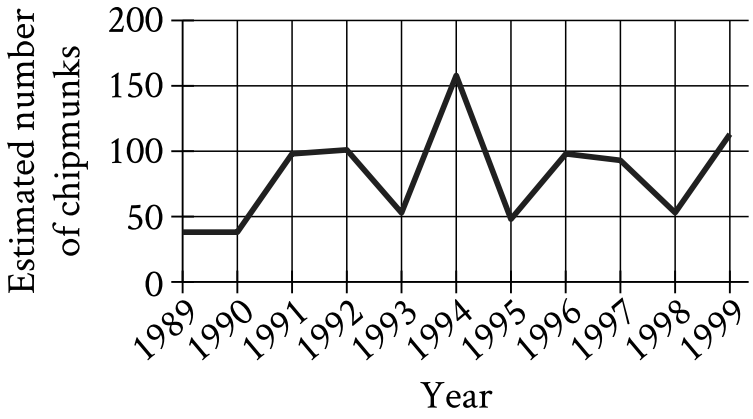
Two-variable data: Models and scatterplots

Problem-Solving and Data Analysis · Math · 71 questions

1 363d5daa-100c-4c65-80bf-bd1ca694191d · Easy

The line graph shows the estimated number of chipmunks in a state park on April 1 of each year from 1989 to 1999.



- The line graph:
  - Begins at 1989, 38 chipmunks
  - Remains level to 1990, 38 chipmunks
  - Rises sharply to 1991, 98 chipmunks
  - Rises gradually to 1992, 101 chipmunks
  - Falls sharply to 1993, 53 chipmunks
  - Rises sharply to 1994, 158 chipmunks
  - Falls sharply to 1995, 48 chipmunks
  - Rises sharply to 1996, 98 chipmunks
  - Falls gradually to 1997, 93 chipmunks
  - Falls sharply to 1998, 53 chipmunks
  - Rises sharply to 1999, 113 chipmunks

Based on the line graph, in which year was the estimated number of chipmunks in the state park the greatest?

A. 1989

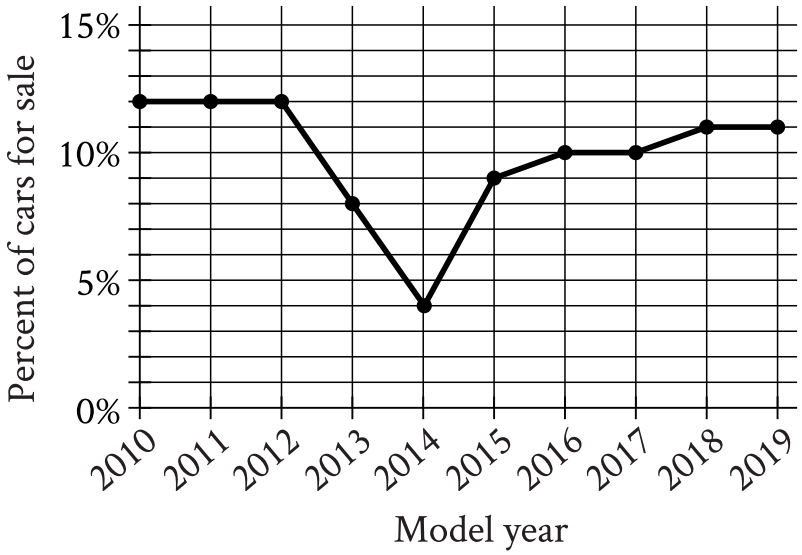
B. 1994

C. 1995

D. 1998

2 41f938b0-0d18-48ac-b9ea-15cc947faaa2 · Easy

The line graph shows the percent of cars for sale at a used car lot on a given day by model year.



- The line graph:
  - Begins at 2010, 12%
  - Remains level to 2011, 12%
  - Remains level to 2012, 12%
  - Falls sharply to 2013, 8%
  - Falls sharply to 2014, 4%
  - Rises sharply to 2015, 9%
  - Rises gradually to 2016, 10%
  - Remains level to 2017, 10%
  - Rises gradually to 2018, 11%
  - Remains level to 2019, 11%

For what model year is the percent of cars for sale the smallest?

A. 2012

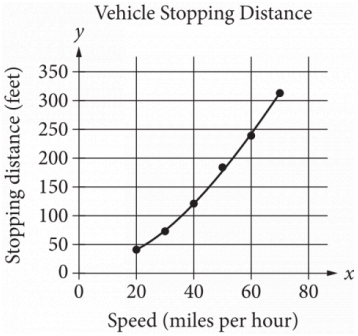
B. 2013

C. 2014

D. 2015

3 01239-DC · Easy

A study was done to determine a new car's stopping distance when it was traveling at different speeds. The study was done on a dry road with good surface conditions. The results are shown below, along with the graph of a quadratic function that models the data.



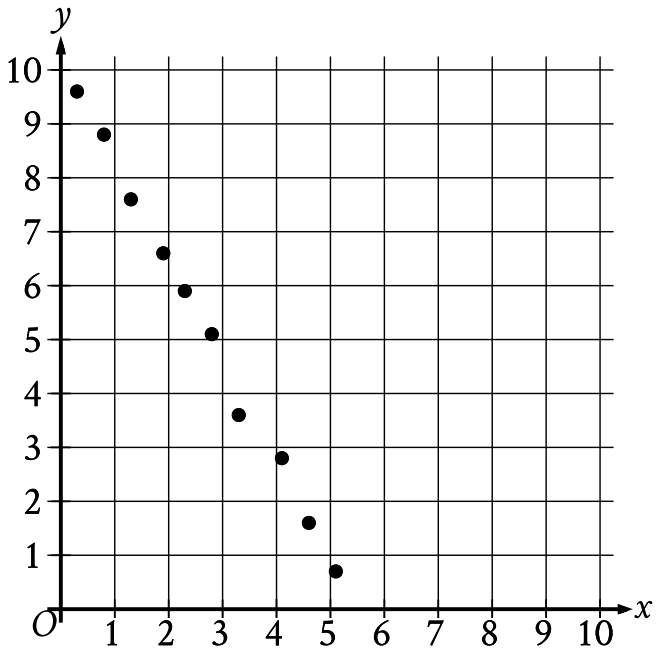
According to the model, which of the following is the best estimate for the stopping distance, in feet, if the vehicle was traveling 55 miles per hour?

A. 25

B. 30

C. 210

D. 250



- The scatterplot has 10 data points.
- The data points are in a linear pattern trending down from left to right.
- Select data points have the following approximate coordinates:
  - (0.8 comma 8.9)
  - (2.8 comma 5.1)
  - (4.1 comma 2.9)

Which of the following equations is the most appropriate linear model for the data shown in the scatterplot?

A. $y = -1.9x - 10.1$

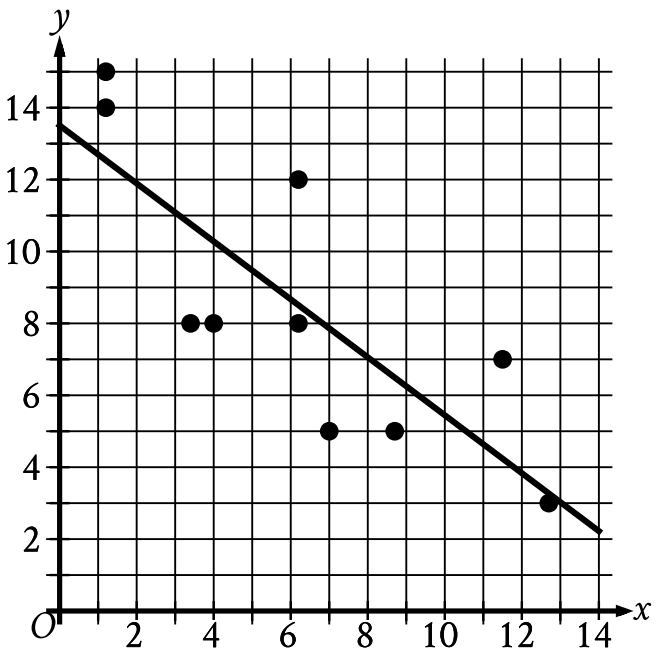
B. $y = -1.9x + 10.1$

C. $y = 1.9x - 10.1$

D. $y = 1.9x + 10.1$

5 77c322ee-efee-4f19-ad6f-03294924b08f · Easy

The scatterplot shows the relationship between two variables, x and y. A line of best fit is also shown.



- The scatterplot has 10 data points.
- The data points are in a linear pattern trending down from left to right.
- A line of best fit is shown:
    - The line of best slants down from left to right.
    - 1 point is touching the line of best fit.
    - 4 points are above the line of best fit.
    - 5 points are below the line of best fit.
    - The line of best fit passes through the following approximate coordinates:
        - (2 comma 12)
        - (8 comma 7)
        - (13 comma 3)

Which of the following equations best represents the line of best fit shown?

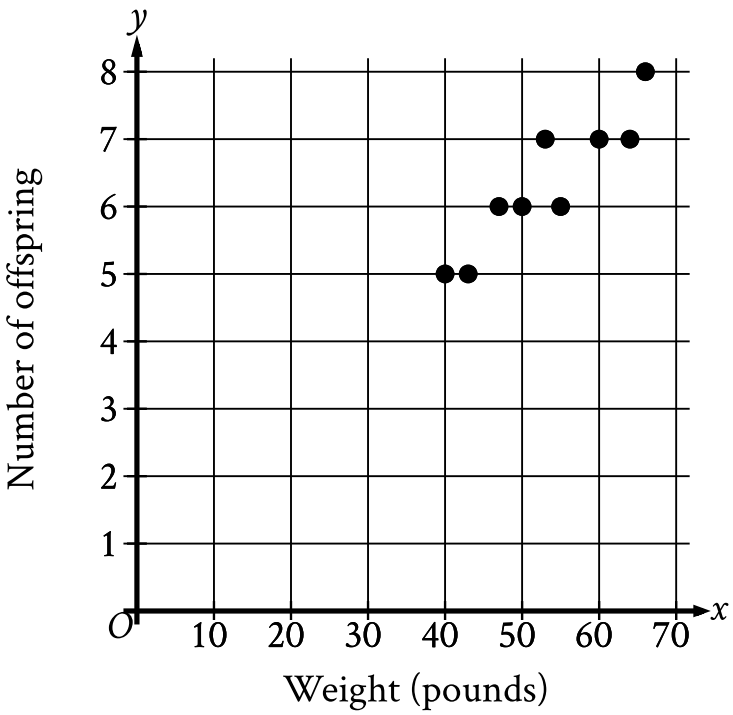
A. $y = 13.5 + 0.8x$

B. $y = 13.5 - 0.8x$

C. $y = -13.5 + 0.8x$

D. $y = -13.5 - 0.8x$

The scatterplot shows the relationship between the weight, in pounds, of each of 9 female gray wolves on April 30 and the number of offspring each gray wolf produced.



- The scatterplot has 9 data points.
- The data points are in a linear pattern trending up from left to right.
- The data points have the following coordinates:
  - (40 comma 5)
  - (43 comma 5)
  - (47 comma 6)
  - (50 comma 6)
  - (53 comma 7)
  - (55 comma 6)
  - (60 comma 7)
  - (64 comma 7)
  - (66 comma 8)

How many offspring did the 50-pound gray wolf produce?

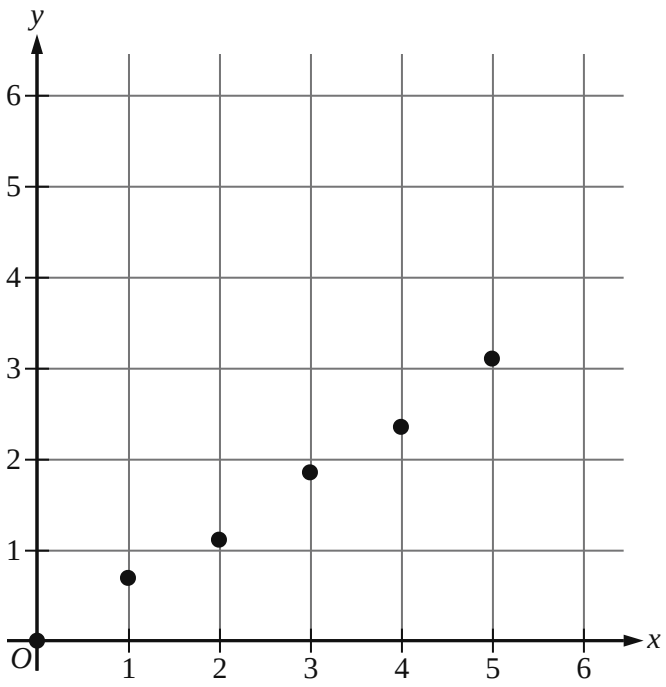
A. 8

B. 7

C. 6

D. 5

7 9fe0c563-5c49-44d6-bd3d-3956ed2de2bd · Easy



- The scatterplot has 6 data points.
- The data points are in a linear pattern trending up from left to right.
- The data points have the following approximate coordinates:
  - (0 comma 0)
  - (1 comma 0.7)
  - (2 comma 1.1)
  - (3 comma 1.9)
  - (4 comma 2.4)
  - (5 comma 3.1)

Which of the following equations is the most appropriate linear model for the data shown in the scatterplot?

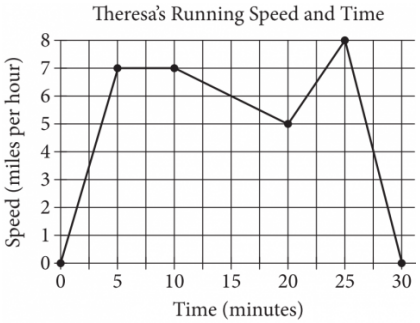
A. $y = -0.606x$

B. $y = 0.606 - 1.001x$

C. $y = 0.606x$

D. $y = 0.606 + 1.001x$

8 07721-DC · Easy



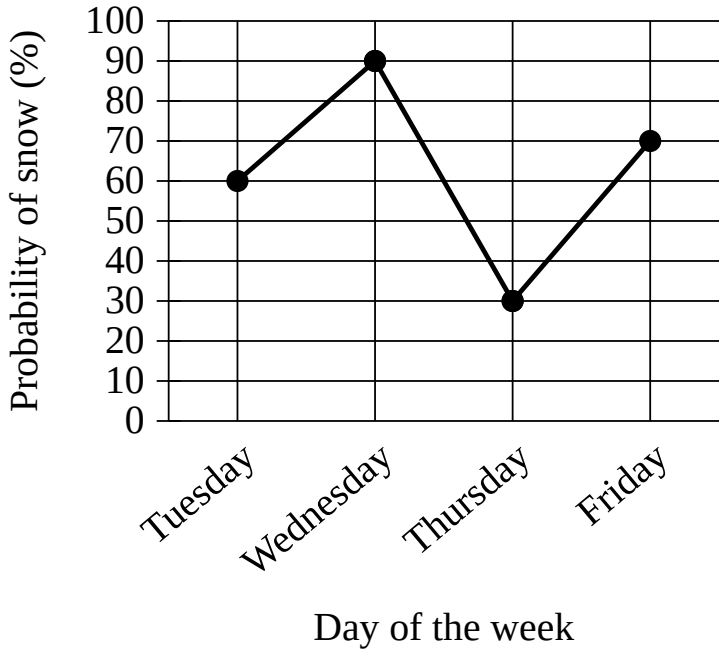
Theresa ran on a treadmill for thirty minutes, and her time and speed are shown on the graph above. According to the graph, which of the following statements is NOT true concerning Theresa's run?

A. Theresa ran at a constant speed for five minutes.

B. Theresa's speed was increasing for a longer period of time than it was decreasing.

C. Theresa's speed decreased at a constant rate during the last five minutes.

D. Theresa's speed reached its maximum during the last ten minutes.



- The line graph:
  - Begins at Tuesday, 60%
  - Rises sharply to Wednesday, 90%
  - Falls sharply to Thursday, 30%
  - Rises sharply to Friday, 70%

The line graph shows the probability of snow, as a percent, at a certain location for each day during a four-day period. According to the line graph, for which day during this four-day period is the probability of snow 30%?

A. Tuesday

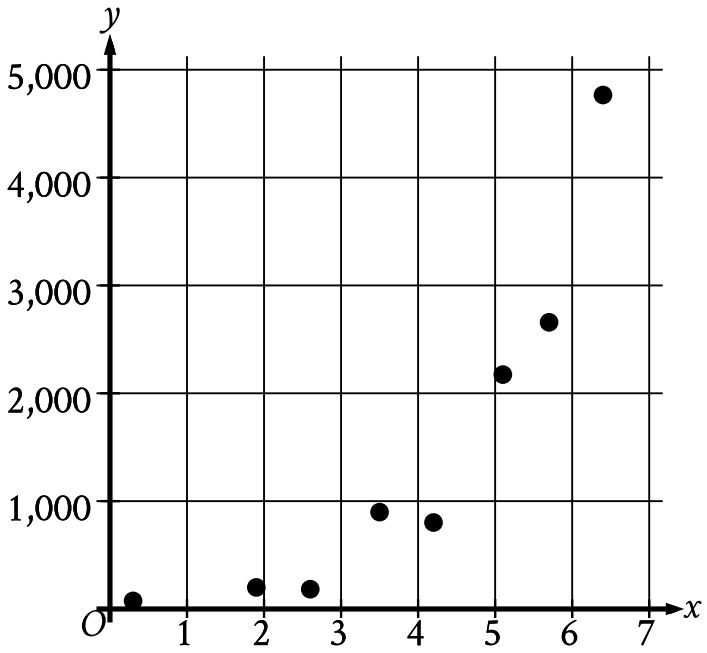
B. Wednesday

C. Thursday

D. Friday

10 aee1cc54-9f2e-46de-94fa-2cccbe708d41 · Easy

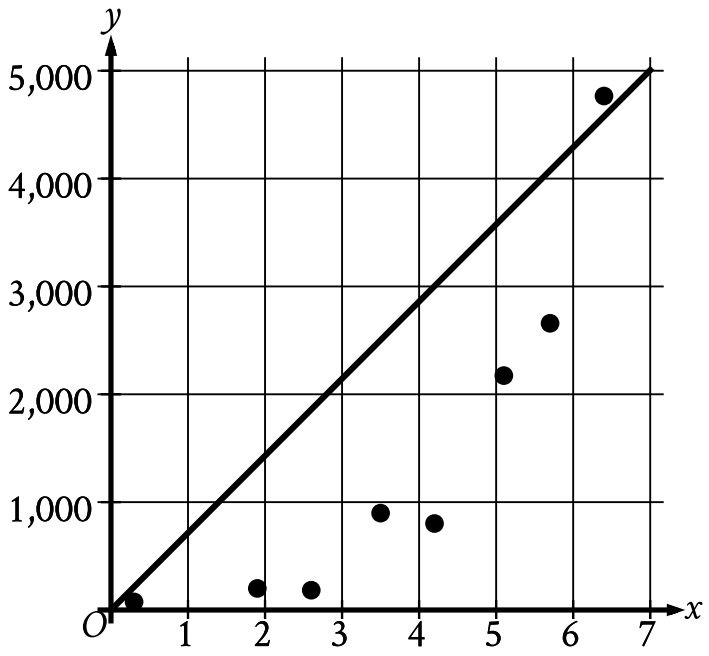
The scatterplot shows the relationship between two variables, x and y.



- The scatterplot has 8 data points.
- The data points are in an exponential pattern trending up from left to right.
- The data points have the following approximate coordinates:
  - (0.3 comma 75)
  - (1.9 comma 200)
  - (2.6 comma 183)
  - (3.5 comma 898)
  - (4.2 comma 802)
  - (5.1 comma 2,173)
  - (5.7 comma 2,658)
  - (6.4 comma 4,766)

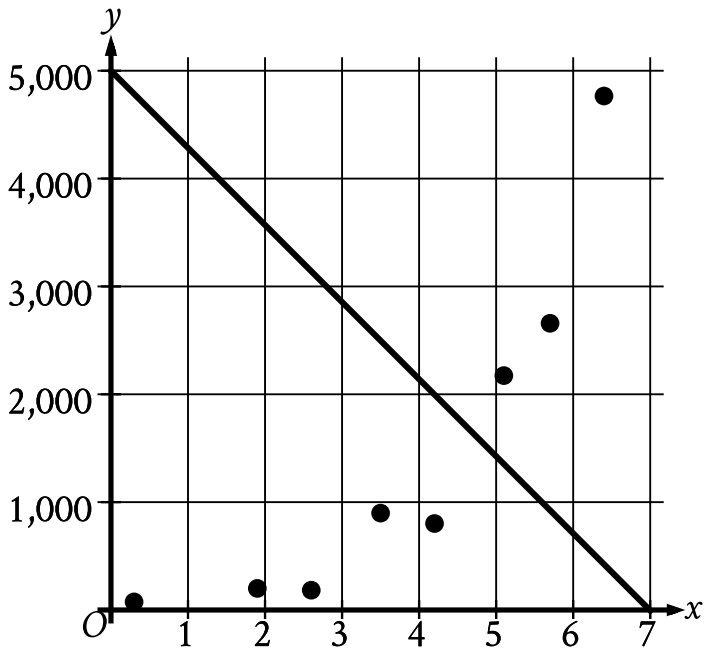
Which of the following graphs shows the most appropriate model for the data?

A.



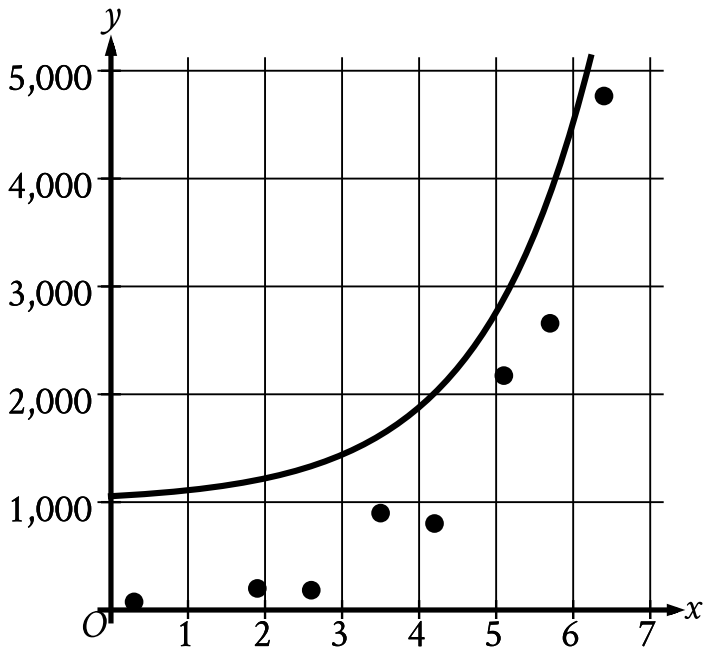
- The scatterplot has 8 data points.
- The data points are in an exponential pattern trending up from left to right.
- A line of best fit is shown:
    - The line of best fit slants up from left to right.
    - 0 points are touching the line of best fit.
    - 1 point is above the line of best fit.
    - 7 points are below the line of best fit.
    - The line of best fit passes through the following approximate coordinates:
        - (0 comma 0)
        - (6 comma 4,290)

B.



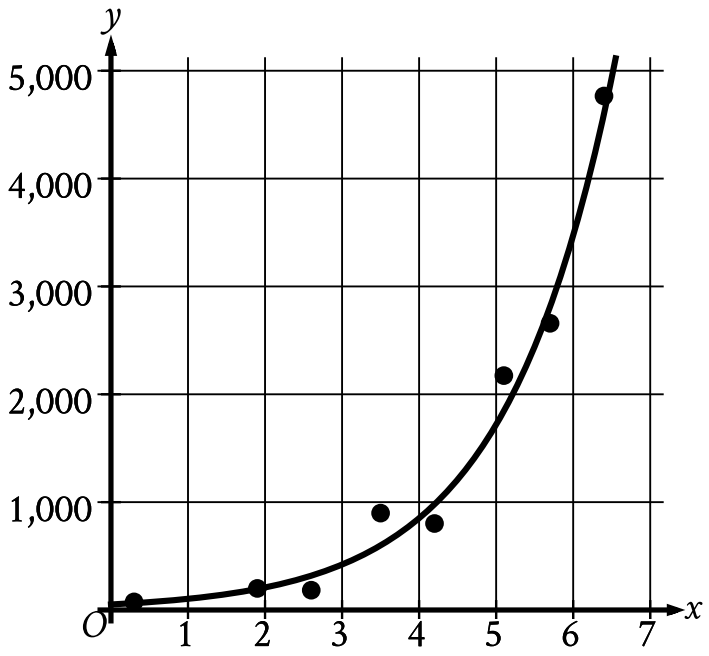
- The scatterplot has 8 data points.
- The data points are in an exponential pattern trending up from left to right.
- A line of best fit is shown:
    - The line of best fit slants down from left to right.
    - 0 points are touching the line of best fit.
    - 3 points are above the line of best fit.
    - 5 points are below the line of best fit.
    - The line of best fit passes through the following coordinates:
        - (0 comma 5,000)
        - (6 comma 710)

C.



- The scatterplot has 8 data points.
- The data points are in an exponential pattern trending up from left to right.
- A curve of best fit is shown:
    - The curve of best fit trends up from left to right.
    - 0 points are touching the curve of best fit.
    - 0 points are above the curve of best fit.
    - 8 points are below the curve of best fit.
    - The curve of best fit passes through the following coordinates:
        - (0 comma 1,055)
        - (4 comma 1,880)
        - (6 comma 4,520)

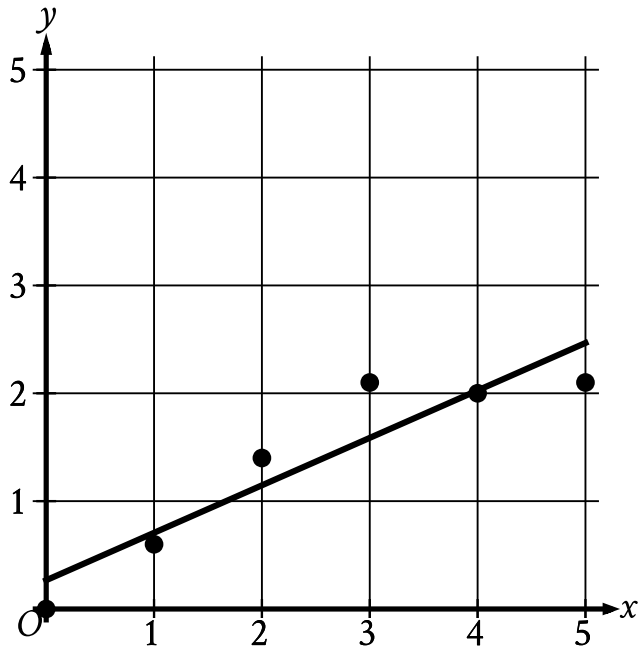
D.



- The scatterplot has 8 data points.
- The data points are in an exponential pattern trending up from left to right.
- A curve of best fit is shown:
    - The curve of best fit trends up from left to right.
    - 2 points are touching the curve of best fit.
    - 3 points are above the curve of best fit.
    - 3 points are below the curve of best fit.
    - The curve of best fit passes through the following approximate coordinates:
        - (0 comma 51)
        - (4 comma 851)
        - (6 comma 3,472)

11 62f651c4-37e7-4fa9-8a3d-ff62ceb23066 · Easy

The scatterplot shows the relationship between x and y. A line of best fit is also shown.



- The scatterplot has 6 data points.
- The data points are in a linear pattern trending up from left to right.
- A line of best fit is shown:
  - The line of best fit slants up from left to right.
  - 1 point is touching the line of best fit.
  - 2 points are above the line of best fit.
  - 3 points are below the lines of best fit.
  - The line of best fit passes through the following approximate coordinates:
    - (0 comma 0.2)
    - (4 comma 2)

Which of the following is closest to the slope of the line of best fit shown?

A. -2.27

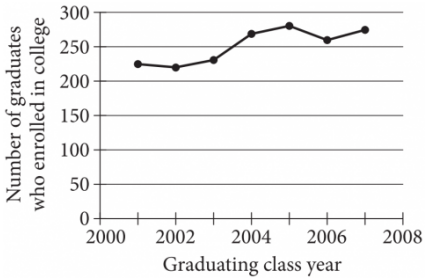
B. -0.44

C. 0.44

D. 2.27

---

12 026230-DC · Easy



The line graph shows the number of graduates from the classes of 2001 through 2007 at a certain school who enrolled in college within 24 months of graduation. Of the following, which class had the fewest graduates who enrolled in college within 24 months of graduation?

A. 2002

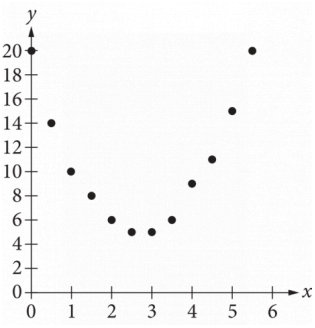
B. 2004

C. 2005

D. 2007

---

13 027703-DC · Easy



Of the following, which is the best model for the data in the scatterplot?

A. $y = 2x^2 - 11x - 20$

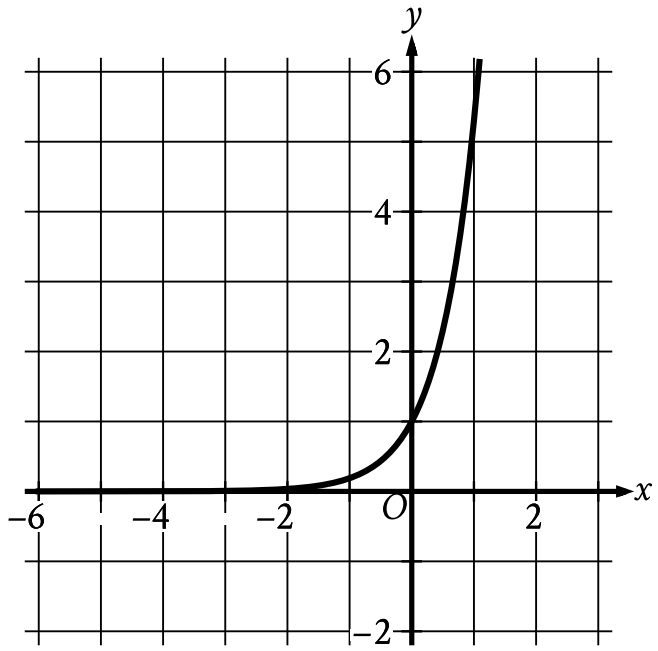
B. $y = 2x^2 - 11x + 20$

C. $y = 2x^2 - 5x - 3$

D. $y = 2x^2 - 5x + 3$

14 3197f818-b46b-411b-ba07-d4d1c811fa29 · Easy

The graph of the function f is shown, where $y = fx$.



- Moving from left to right:
  - The curve passes from quadrant 2 to quadrant 1.
  - In quadrant 2, the curve trends up gradually to point (0 comma 1).
  - In quadrant 1, the curve trends up sharply.
- As x decreases, the curve approaches the line y equals 0.
- The curve passes through the following points:
  - approximately (negative 1 comma 0.2)
  - (0 comma 1)
  - approximately (1 comma 5.3)

Which of the following best describes the function f?

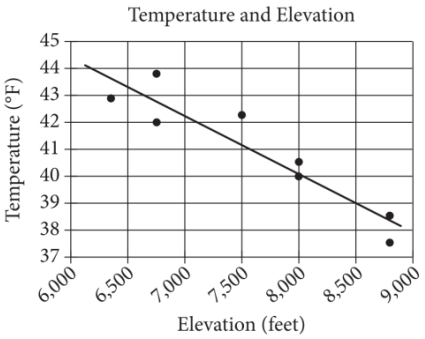
A. Decreasing exponential

B. Increasing exponential

C. Decreasing linear

D. Increasing linear

15 070913-DC · Easy



The scatterplot above shows the high temperature on a certain day and the elevation of 8 different locations in the Lake Tahoe Basin. A line of best fit for the data is also shown. Which of the following statements best describes the association between the elevation and the temperature of locations in the Lake Tahoe Basin?

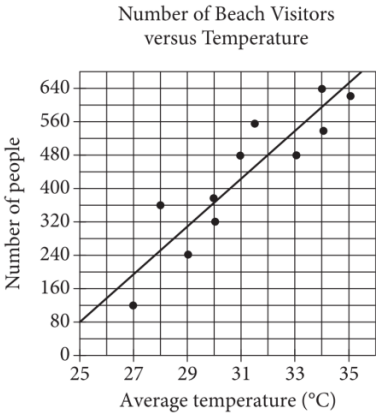
A. As the elevation increases, the temperature tends to increase.

B. As the elevation increases, the temperature tends to decrease.

C. As the elevation decreases, the temperature tends to decrease.

D. There is no association between the elevation and the temperature.

16 070917-DC · Easy

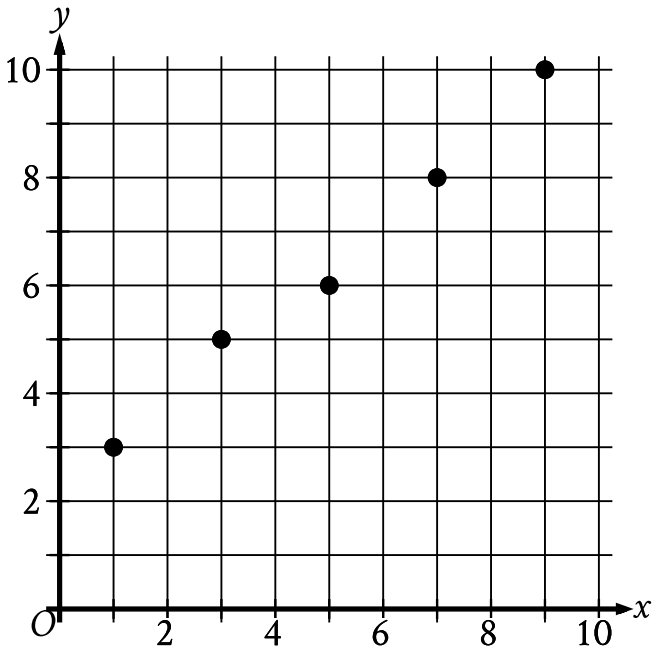


Each dot in the scatterplot above represents the temperature and the number of people who visited a beach in Lagos, Nigeria, on one of eleven different days. The line of best fit for the data is also shown. According to the line of best fit, what is the number of people, rounded to the nearest 10, predicted to visit this beach on a day with an average temperature of 32°C?

Student-produced response: _____

17 69235d8f-a27b-49a1-a4e9-e905f8b73ef4 · Easy

The scatterplot shows the relationship between two variables, x and y.



- The scatterplot has 5 data points.
- The data points are in a linear pattern trending up from left to right.
- The data points have the following coordinates:
  - (1 comma 3.0)
  - (3 comma 5.0)
  - (5 comma 6.0)
  - (7 comma 8.0)
  - (9 comma 10.0)

Which equation is the most appropriate linear model for this relationship?

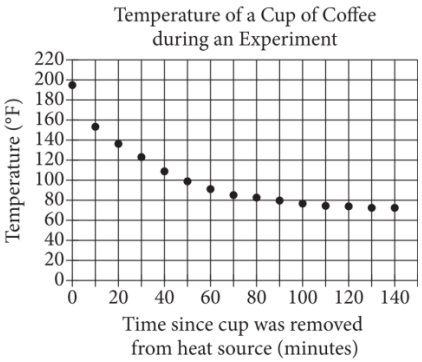
A. $y = -0.9x - 2.2$

B. $y = -0.9x + 2.2$

C. $y = -0.9x$

D. $y = 0.9x + 2.2$

18 070915-DC · Easy



In an experiment, a heated cup of coffee is removed from a heat source, and the cup of coffee is then left in a room that is kept at a constant temperature. The graph above shows the temperature, in degrees Fahrenheit (°F), of the coffee immediately after being removed from the heat source and at 10-minute intervals thereafter. During which of the following 10-minute intervals does the temperature of the coffee decrease at the greatest average rate?

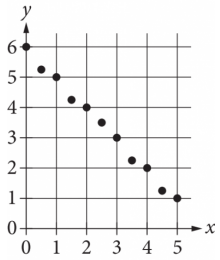
A. Between 0 and 10 minutes

B. Between 30 and 40 minutes

C. Between 50 and 60 minutes

D. Between 90 and 100 minutes

19 027702-DC · Easy



Which of the following could be an equation for a line of best fit for the data in the scatterplot?

A. $y = -x + 6$

B. $y = -x - 6$

C. $y = 6x + 1$

D. $y = 6x - 1$

---

20 c15291f0-3442-4787-9e24-c2044f3e1f6f · Easy

An airplane descends from an altitude of $9{,}500$ feet to $5{,}000$ feet at a constant rate of 400 feet per minute. What type of function best models the relationship between the descending airplane's altitude and time?

A. Decreasing exponential

B. Decreasing linear

C. Increasing exponential

D. Increasing linear

---

21 4c22545c-b1dc-417e-a342-b8115c476ce0 · Easy

The table shows selected values from function *f*.

x	*fx*
-1	16
0	17
1	18
2	19

Which of the following is the best description of function *f*?

A. Decreasing linear

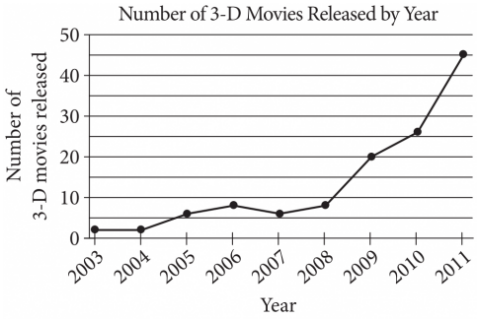
B. Increasing linear

C. Decreasing exponential

D. Increasing exponential

---

22 07714-DC · Easy



According to the line graph above, between which two consecutive years was there the greatest change in the number of 3-D movies released?

A. 2003–2004

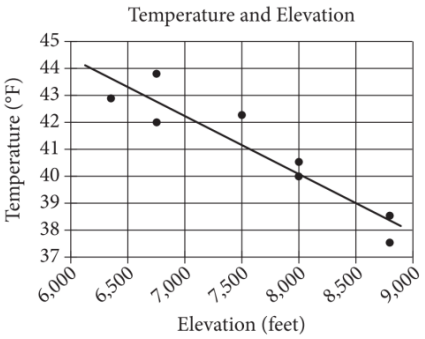
B. 2008–2009

C. 2009–2010

D. 2010–2011

---

23 070916-DC · Easy



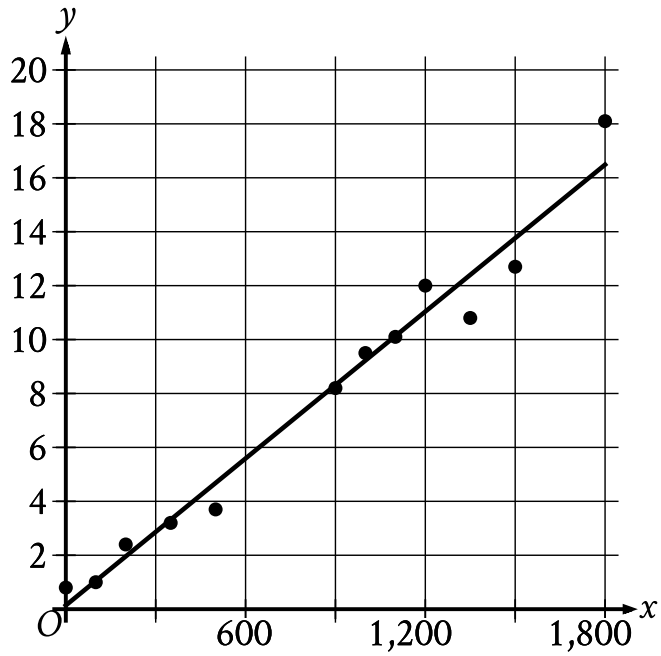
The scatterplot above shows the high temperature on a certain day and the elevation of 8 different locations in the Lake Tahoe Basin. A line of best fit for the data is also shown. What temperature is predicted by the line of best fit for a location in the Lake Tahoe Basin with an elevation of 8,500 feet?

A. 37°F

B. 39°F

C. 41°F

D. 43°F



- The scatterplot has 12 data points.
- The data points are in a linear pattern trending up from left to right.
- A line of best fit is shown:
    - The line of best fit slants up from left to right.
    - The line of best fit passes through the following approximate coordinates:
        - (600 comma 5.6)
        - (1,200 comma 11.0)
        - (1,500 comma 13.8)
        - (1,800 comma 16.5)

Twelve data points are shown in the scatterplot. A line of best fit for the data is also shown. At $x = 1{,}200$, which of the following is closest to the y-value predicted by the line of best fit?

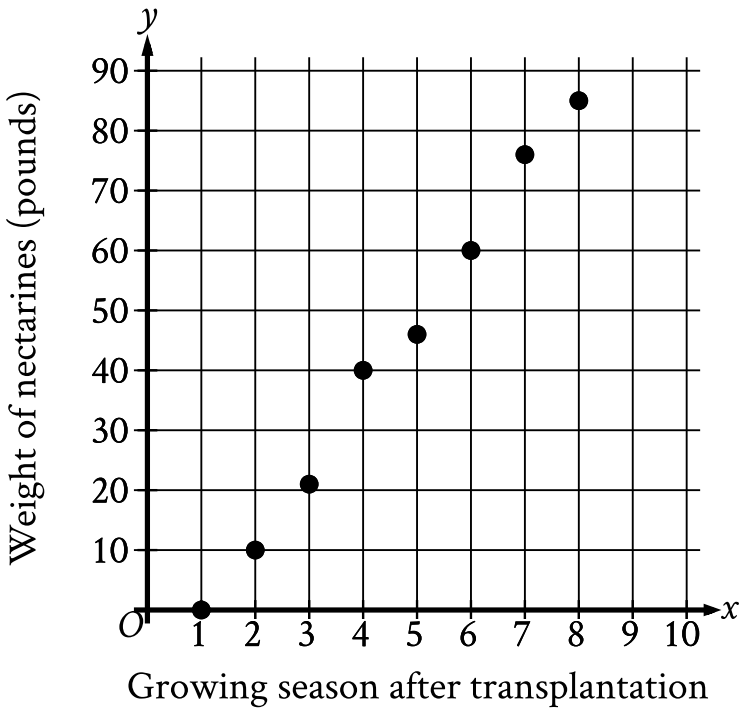
A. 16

B. 14

C. 11

D. 6

An orchard owner recorded the weight, in pounds, of all nectarines that grew on a dwarf nectarine tree during each growing season after the tree's transplantation. The scatterplot shows this weight, in pounds, for each growing season after the tree's transplantation.



- The scatterplot has 8 data points.
- The data points are in a linear pattern trending up from left to right.
- The data points have the following coordinates:
  - (1 comma 0)
  - (2 comma 10)
  - (3 comma 21)
  - (4 comma 40)
  - (5 comma 46)
  - (6 comma 60)
  - (7 comma 76)
  - (8 comma 85)

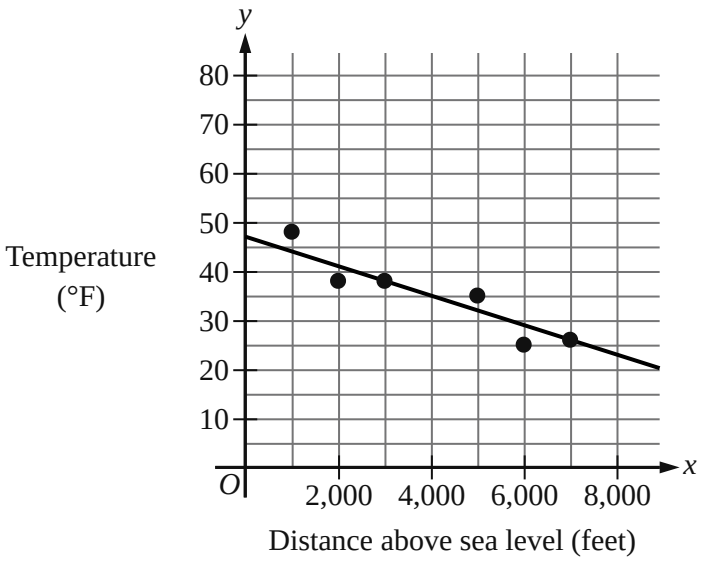
What was the weight, to the nearest pound, of all nectarines that grew on the tree during the 4th growing season after the tree's transplantation?

Student-produced response: _____

---

26 be404513-86c1-473f-9573-d0ef7f6ac666 · Easy

The scatterplot shows the temperature, in degrees Fahrenheit °F, and the distance above sea level, in feet, measured at 6 locations on Mount Jefferson. A line of best fit is also shown.



- The scatterplot has 6 data points.
- The data points are in a linear pattern trending down from left to right.
- A line of best fit is shown:
  - The line of best fit slants down from left to right.
  - The line of best fit passes through the following coordinates:
    - (0 comma 47)
    - (4,000 comma 35)
    - (7,000 comma 26)

At a distance of 4,000 feet above sea level, what is the temperature, in °F, predicted by the line of best fit?

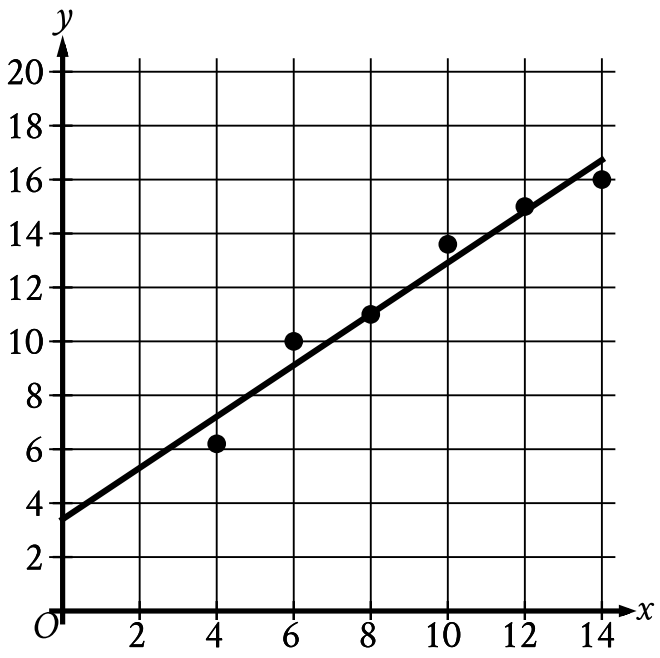
A. 47

B. 35

C. 25

D. 0

The scatterplot shows the relationship between two variables, x and y. A line of best fit is also shown.



- The scatterplot has 6 data points.
- The data points are in a linear pattern trending up from left to right.
- A line of best fit is shown:
    - The line of best fit slants up from left to right.
    - The line of best fit passes through the following approximate coordinates:
        - (0 comma 3.4)
        - (8 comma 11)
        - (12 comma 14.8)

Which of the following equations best represents the line of best fit shown?

A. $y = x + 3.4$

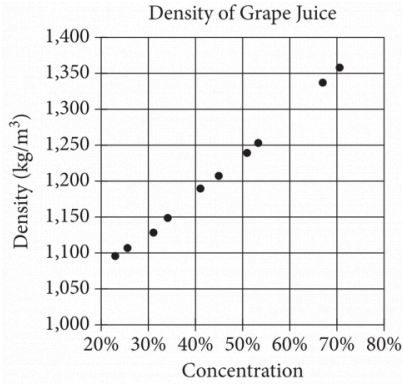
B. $y = x - 3.4$

C. $y = -x + 3.4$

D. $y = -x - 3.4$

---

28 031645-DC · Easy



The densities of different concentrations of grape juice are shown in the scatterplot above. According to the trend shown by the data, which of the following is closest to the predicted density, in kilograms per cubic meter (kg/m^3), for grape juice with a concentration of 60%?

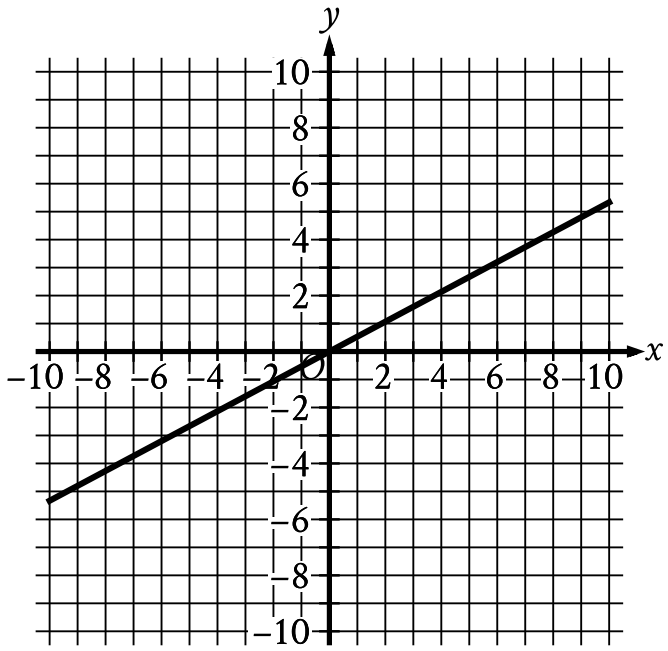
A. 1,200

B. 1,250

C. 1,300

D. 1,350

The graph of function f is shown, where $y = fx$.



- The line slants gradually up from left to right.
- The line passes through the following points:
  - (negative 6 comma negative StartFraction 16 Over 5 EndFraction)
  - (0 comma 0)
  - (6 comma StartFraction 16 Over 5 EndFraction)

Which of the following describes function f?

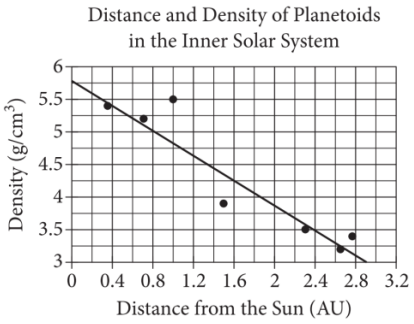
A. Increasing linear

B. Decreasing linear

C. Increasing exponential

D. Decreasing exponential

30 070914-DC · Easy



The scatterplot above shows the densities of 7 planetoids, in grams per cubic centimeter, with respect to their average distances from the Sun in astronomical units (AU). The line of best fit is also shown. An astronomer has discovered a new planetoid about 1.2 AU from the Sun. According to the line of best fit, which of the following best approximates the density of the planetoid, in grams per cubic centimeter?

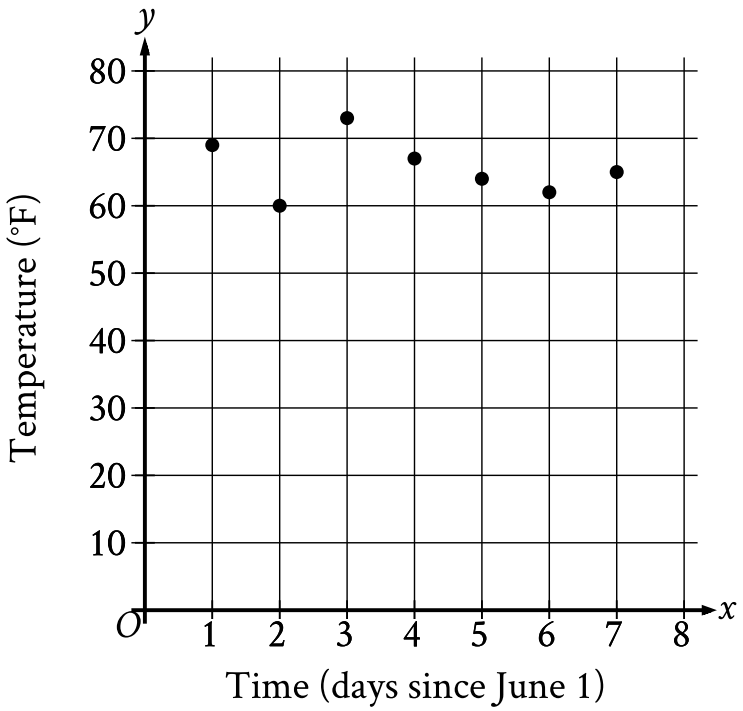
A. 3.6

B. 4.1

C. 4.6

D. 5.5

The scatterplot shows the temperature y, in °F, recorded by a meteorologist at various times x, in days since June 1.



- The scatterplot has 7 data points.
- The data points are in a linear pattern trending approximately horizontally.
- The data points have the following coordinates:
  - (1 comma 69)
  - (2 comma 60)
  - (3 comma 73)
  - (4 comma 67)
  - (5 comma 64)
  - (6 comma 62)
  - (7 comma 65)

During which of the following time periods did the greatest increase in recorded temperature take place?

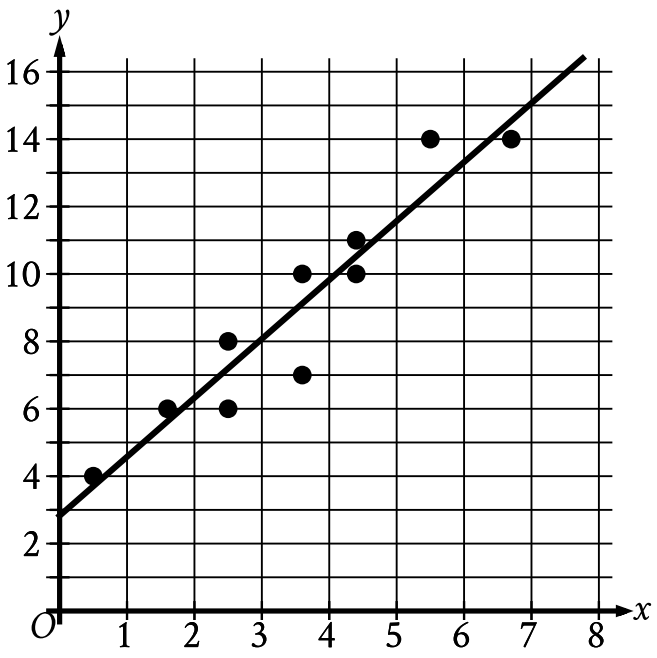
A. From $x = 6$ to $x = 7$

B. From $x = 5$ to $x = 6$

C. From $x = 2$ to $x = 3$

D. From $x = 1$ to $x = 2$

The scatterplot shows the relationship between two variables, x and y. A line of best fit is also shown.



- The scatterplot has 10 data points.
- The data points are in a linear pattern trending up from left to right.
- A line of best fit is shown.
    - The line of best fit slants up from left to right.
    - 1 point is touching the line of best fit.
    - 5 points are above the line of best fit.
    - 4 points are below the line of best fit.
    - The line of best fit passes through the following approximate coordinates:
        - (0 comma 3)
        - (3 comma 8)
        - (7 comma 15)

Which of the following equations best represents the line of best fit shown?

A. $y = 2.8 + 1.7x$

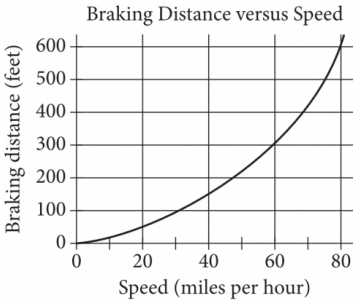
B. $y = 2.8 - 1.7x$

C. $y = -2.8 + 1.7x$

D. $y = -2.8 - 1.7x$

---

33 031603-DC · Easy



The graph above shows the relationship between the speed of a particular car, in miles per hour, and its corresponding braking distance, in feet. Approximately how many feet greater will the car's braking distance be when the car is traveling at 50 miles per hour than when the car is traveling at 30 miles per hour?

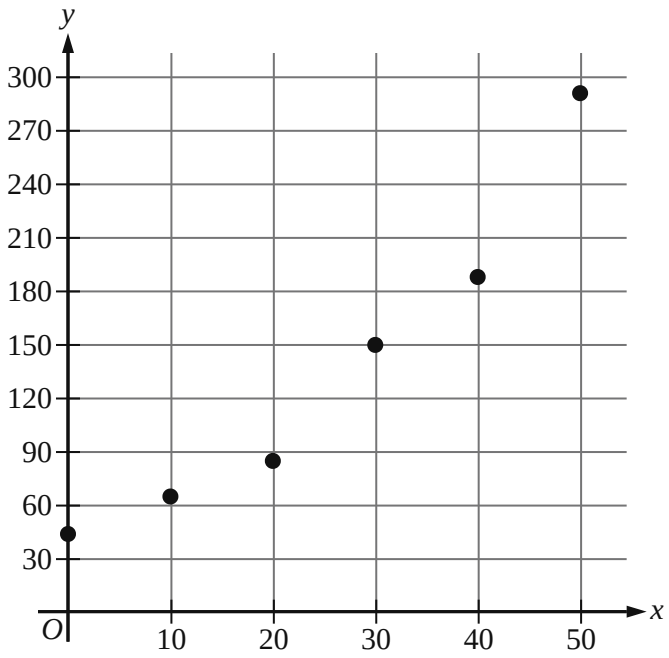
A. 75

B. 125

C. 175

D. 250

The scatterplot shows the relationship between the number of swans, y, in a certain population and the number of years since 1970, x, where $0 \leq x \leq 50$.



- The scatterplot has 6 data points.
- The data points are in an exponential pattern trending up from left to right.
- The data points have the following approximate coordinates:
  - (0 comma 43.53)
  - (10 comma 64.53)
  - (20 comma 84.47)
  - (30 comma 149.47)
  - (40 comma 187.53)
  - (50 comma 290.53)

Which of the following equations is the most appropriate exponential model for the data shown in the scatterplot?

A. $y = 431.04^{-x}$

B. $y = 431.04^{x}$

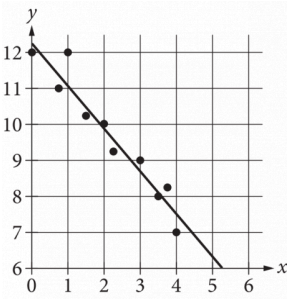
C. $y = 2881.04^{-x}$

D. $y = 2881.04^x$

---

35 027709-DC · Medium

The scatterplot shows the relationship between two variables, x and y. A line of best fit for the data is also shown. Which of the following is closest to the difference between the y-coordinate of the data point with $x = 1$ and the y-value predicted by the line of best fit at $x = 1$?



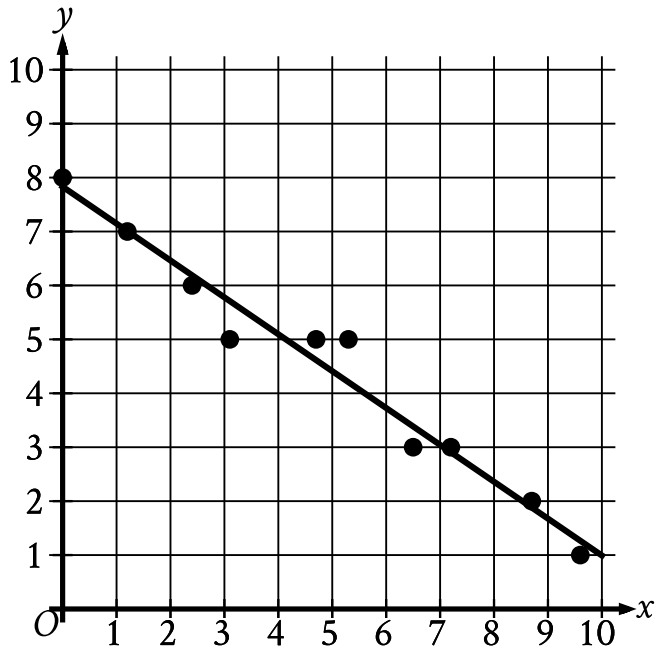
A. 1

B. 2

C. 5

D. 12

In the given scatterplot, a line of best fit for the data is shown.



- The scatterplot has 10 data points.
- The data points are in a linear pattern trending down from left to right.
- A line of best fit is shown:
- The line of best fit slants down from left to right.
    - 5 points are touching the line of best fit.
    - 2 points are above the line of best fit.
    - 3 points are below the line of best fit.
    - The line of best fit passes through the following approximate coordinates:
        - (1 comma 7)
        - (4 comma 5)
        - (10 comma 1)

Which of the following is closest to the slope of this line of best fit?

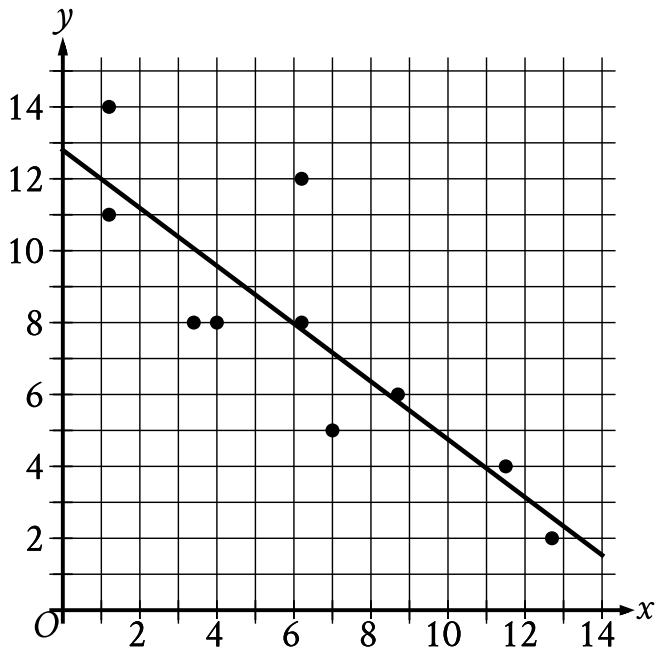
A. 7

B. 0.7

C. -0.7

D. -7

The scatterplot shows the relationship between two variables, x and y. A line of best fit is also shown.



- The scatterplot has 10 data points.
- The data points are in a linear pattern trending down from left to right.
- A line of best fit is shown:
    - The line of best fit slants down from left to right.
    - 2 points are touching the line of best fit.
    - 3 points are above the line of best fit.
    - 5 points are below the line of best fit.
    - The line of best fit passes through the following approximate coordinates:
        - (0 comma 12.9)
        - (6 comma 8)
        - (12 comma 3.2)

Which of the following is closest to the slope of the line of best fit shown?

A. -2.4

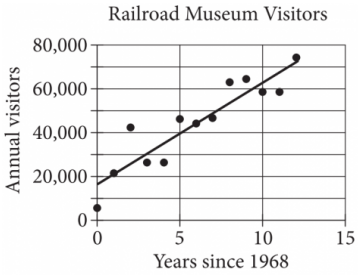
B. -0.8

C. 0.8

D. 2.4

---

38 026150-DC · Medium



The scatterplot above shows the number of visitors to a railroad museum in Pennsylvania each year from 1968 to 1980, where t is the number of years since 1968 and n is the number of visitors. A line of best fit is also shown. Which of the following could be an equation of the line of best fit shown?

A. $n = 16{,}090 + 4{,}680t$

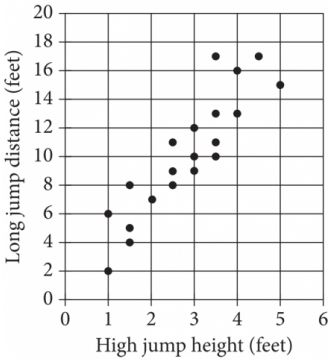
B. $n = 4{,}690 + 16{,}090t$

C. $n = 16{,}090 + 9{,}060t$

D. $n = 9{,}060 + 16{,}090t$

---

39 028872-DC · Medium



Each dot in the scatterplot above represents the height x, in feet, in the high jump, and the distance y, in feet, in the long jump, made by each student in a group of twenty students. The graph of which of the following equations is a line that most closely fits the data?

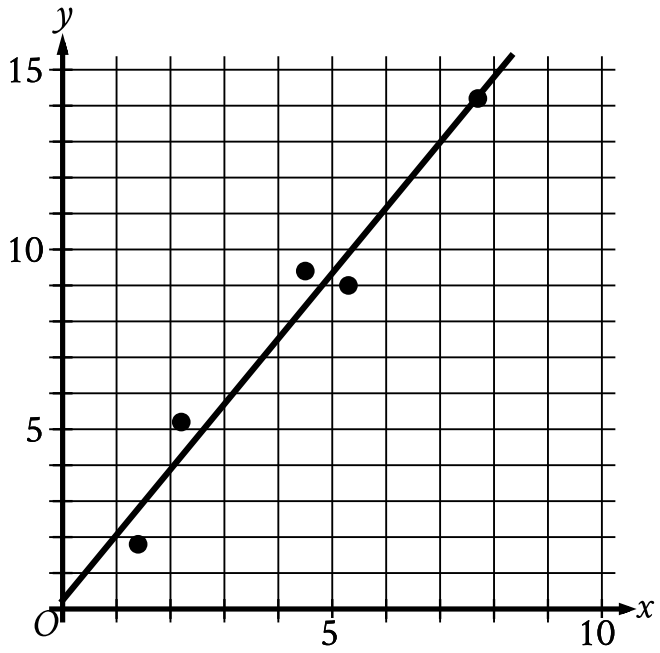
A. $y = 0.82x + 3.30$

B. $y = 0.82x - 0.82$

C. $y = 3.30x + 0.82$

D. $y = 3.30x - 3.30$

In the given scatterplot, a line of best fit for the data is shown.



- The scatterplot has 5 data points.
- The data points are in a linear pattern trending up from left to right.
- A line of best fit is shown:
  - The line of best fit slants up from left to right.
  - The line of best fit passes through the following approximate coordinates:
    - (0 comma 0.2)
    - (5 comma 9.3)

Which of the following is closest to the slope of the line of best fit shown?

A. 0.2

B. 0.7

C. 1.8

D. 2.6

41 11102-DC · Medium

An inspector begins a day of work with a large sample of shirts that need to be checked for defects. The inspector works at a constant rate throughout the morning. What type of model is best to model the number of shirts remaining to be checked for defects at any given time throughout the morning?

A. A linear model with a positive slope

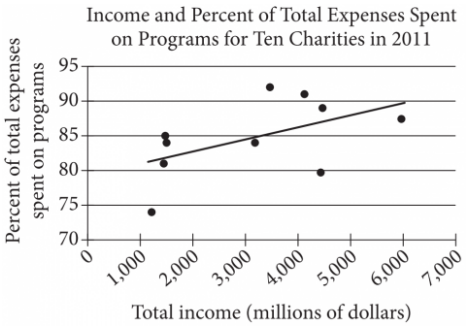
B. A linear model with a negative slope

C. An exponential growth model

D. An exponential decay model

---

42 08424-DC · Medium



The scatterplot above shows data for ten charities along with the line of best fit. For the charity with the greatest percent of total expenses spent on programs, which of the following is closest to the difference of the actual percent and the percent predicted by the line of best fit?

A. 10%

B. 7%

C. 4%

D. 1%

---

43 031669-DC · Medium

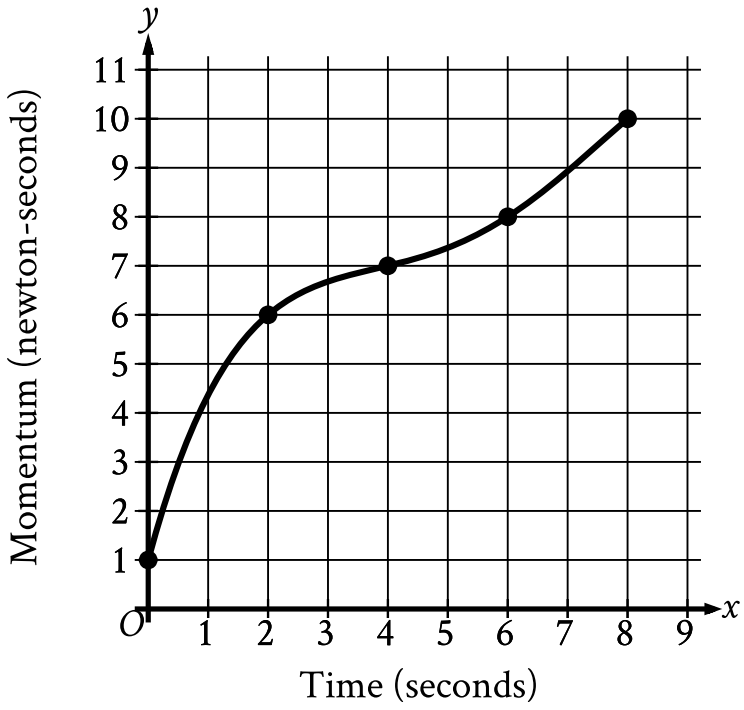
Which of the following is true about the values of 2^x and $2x+2$ for $x>0$?

A. For all $x>0$, it is true that $2^x<2x+2$.

B. For all $x>0$, it is true that $2^x>2x+2$.

C. There is a constant c such that if $0<x<c$, then $2^x<2x+2$, but if $x>c$, then $2^x>2x+2$.

D. There is a constant c such that if $0<x<c$, then $2^x>2x+2$, but if $x>c$, then $2^x<2x+2$.

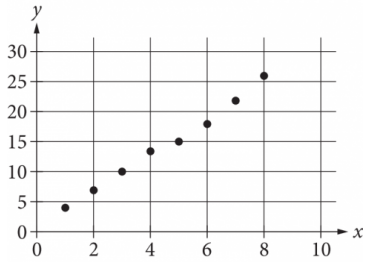


- In quadrant 1:
  - The curve begins at point (0 comma 1).
  - The curve rises sharply to point (2 comma 6).
  - The curve rises gradually to point (4 comma 7).
  - The curve rises gradually to point (6 comma 8).
  - The curve rises sharply and ends at point (8 comma 10).

The graph shows the momentum y, in newton-seconds, of an object x seconds after the object started moving, for $0 \leq x \leq 8$. What is the average rate of change, in newton-seconds per second, in the momentum of the object from $x = 2$ to $x = 6$?

Student-produced response: _____

45 026242-DC · Medium



Which of the following could be the equation for a line of best fit for the data shown in the scatterplot above?

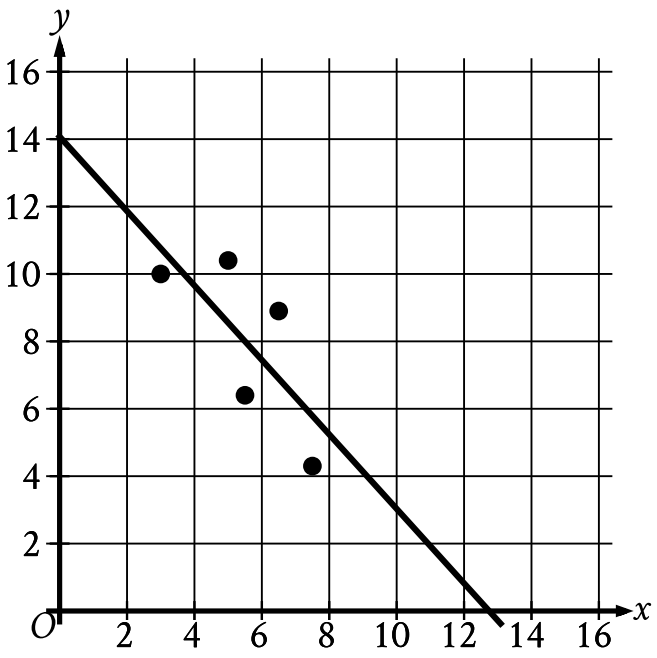
A. $y = 3x + 0.8$

B. $y = 0.8x + 3$

C. $y = -0.8x + 3$

D. $y = -3x + 0.8$

The scatterplot shows the relationship between two variables, x and y. A line of best fit is also shown.



- The scatterplot has 5 data points.
- The data points are in a linear pattern trending down from left to right.
- A line of best fit is shown:
    - The line of best fit slants down from left to right.
    - The line of best fit passes through the following approximate coordinates:
        - (0 comma 14.1)
        - (12.7 comma 0)

Which of the following is closest to the slope of this line of best fit?

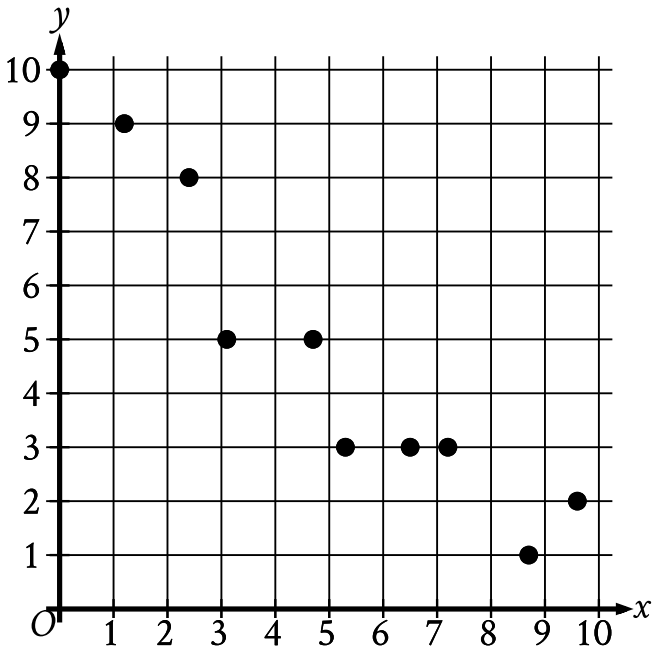
A. -3.3

B. -1.1

C. 1.1

D. 3.3

The scatterplot shows the relationship between two variables, x and y.



- The scatterplot has 10 data points.
- The data points are in a linear pattern trending down from left to right.
- The data points have the following approximate coordinates:
  - (0 comma 10)
  - (1.2 comma 9)
  - (2.3 comma 8)
  - (3.1 comma 5)
  - (4.8 comma 5)
  - (5.2 comma 3)
  - (6.5 comma 3)
  - (7.2 comma 3)
  - (9.6 comma 2)
  - (8.8 comma 1)

Which of the following equations is the most appropriate linear model for the data shown?

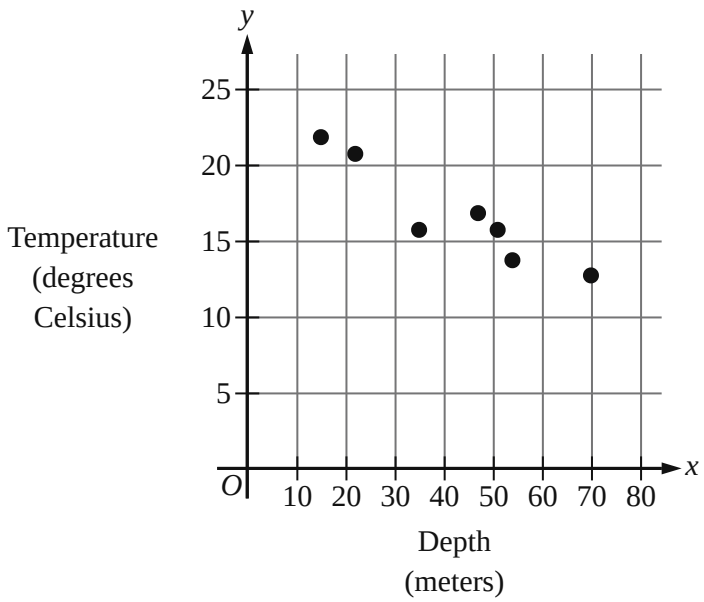
A. $y = 0.9 + 9.4x$

B. $y = 0.9 - 9.4x$

C. $y = 9.4 + 0.9x$

D. $y = 9.4 - 0.9x$

As part of a study of lake conditions, the water temperature was recorded at different depths below the surface of a certain lake. The data shown in the scatterplot give the recorded temperature, in degrees Celsius, for 7 depths, in meters, below the surface of the lake.



- The scatterplot has 7 data points.
- The data points are in a linear pattern trending down from left to right.
- The data points have the following approximate coordinates:
  - (15 comma 21.80)
  - (22 comma 20.70)
  - (35 comma 15.70)
  - (47 comma 16.80)
  - (51 comma 15.70)
  - (54 comma 13.70)
  - (70 comma 12.70)

Which of the following is closest to the slope of a line of best fit for the data shown?

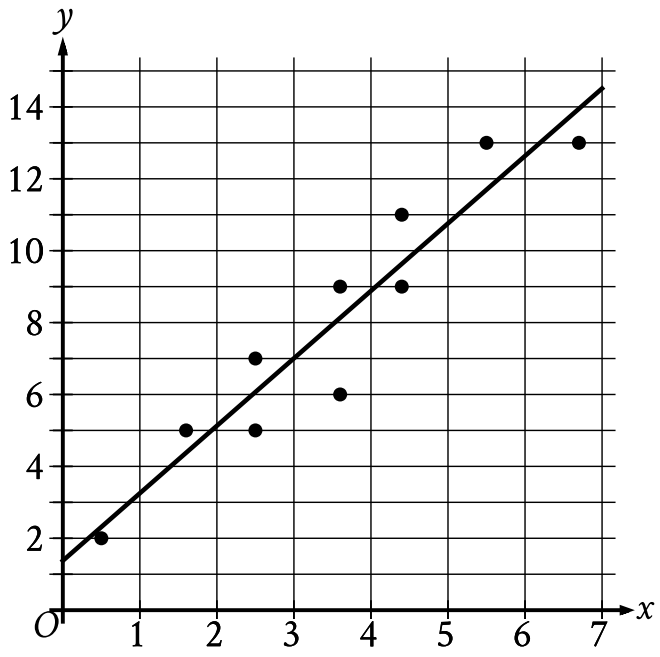
A. -0.17

B. -2.17

C. -6.02

D. -8.02

In the given scatterplot, a line of best fit for the data is shown.



- The scatterplot has 10 data points.
- The data points are in a linear pattern trending up from left to right.
- A line of best fit is shown:
    - The line of best fit slants up from left to right.
    - The line of best fit passes through the following approximate coordinates:
        - (1 comma 3.3)
        - (3 comma 7.0)
        - (5 comma 10.8)

Which of the following is closest to the slope of the line of best fit shown?

A. 0

B. $\frac{1}{2}$

C. 1

D. 2

50 741f638a-272e-4982-91a9-50a01668f587 · Medium

Each year, the value of an investment increases by 0.49% of its value the previous year. Which of the following functions best models how the value of the investment changes over time?

A. Decreasing exponential

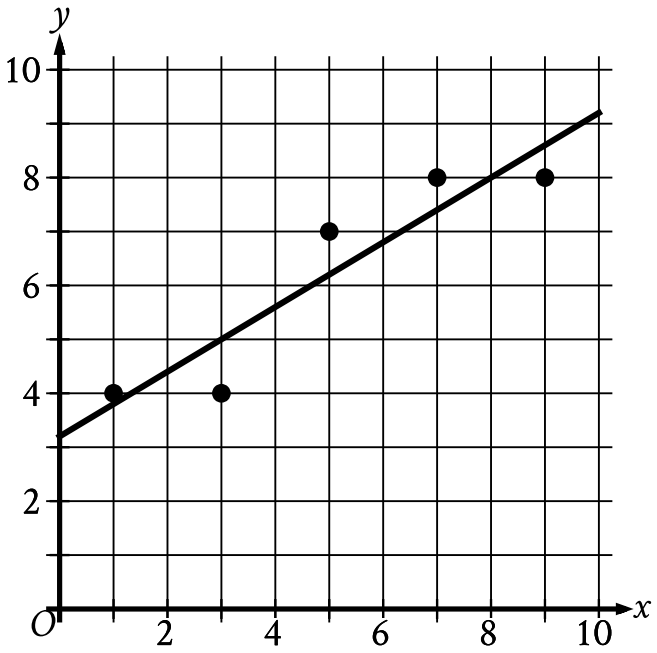
B. Decreasing linear

C. Increasing exponential

D. Increasing linear

51 76b9ab2a-a161-4577-8b1b-7e587c2a4543 · Medium

The scatterplot shows the relationship between x and y. A line of best fit is also shown.



- The scatterplot has 5 data points.
- The data points are in a linear pattern trending up from left to right.
- A line of best fit is shown:
  - The line of best fit slants up from left to right.
  - The line of best fit passes through the following approximate coordinates:
    - (0 comma 3.20)
    - (1 comma 3.80)
    - (2 comma 4.40)

Which of the following is closest to the slope of this line of best fit?

A. 0.60

B. 2.50

C. 7.80

D. 8.00

52 028832-DC · Medium

In which of the following tables is the relationship between the values of x and their corresponding y-values nonlinear?

A.

x	1	2	3	4
y	8	11	14	17

B.

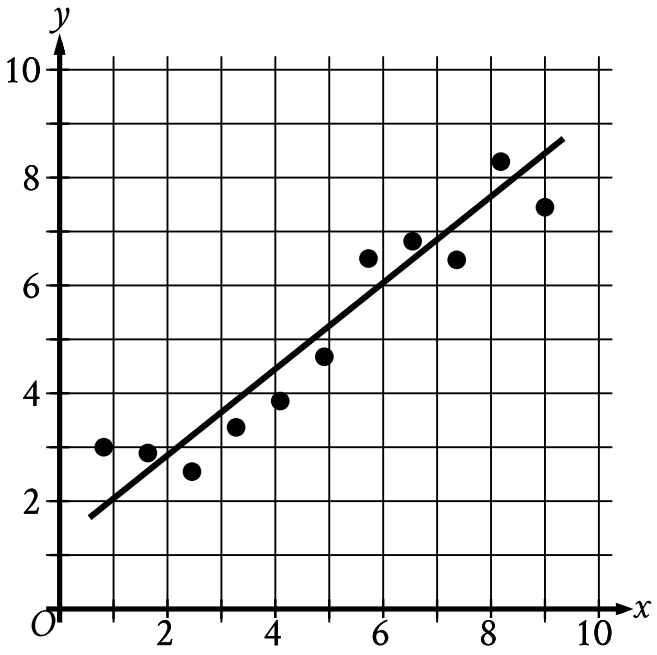
x	1	2	3	4
y	4	8	12	16

C.

x	1	2	3	4
y	8	13	18	23

D.

x	1	2	3	4
y	6	12	24	48

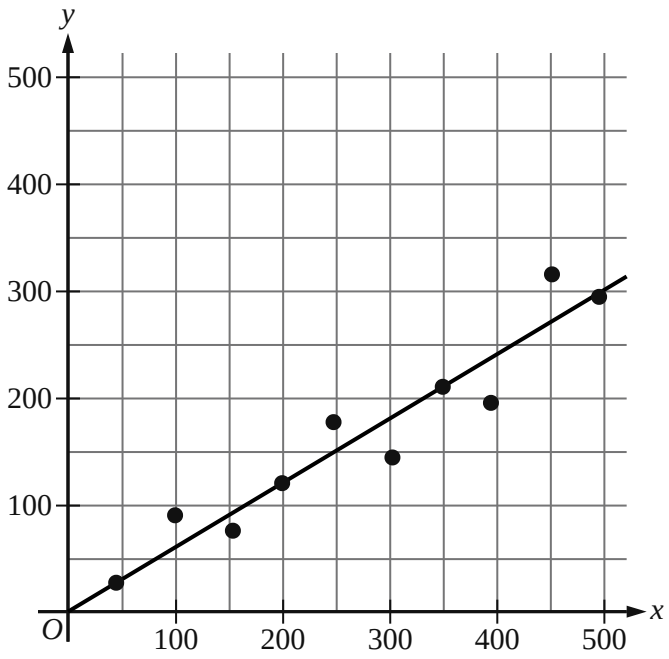


- The scatterplot has 11 data points.
- The data points are in a linear pattern trending up from left to right.
- A line of best fit is shown:
  - The line of best fit slants up from left to right.
  - 5 points are above the line of best fit.
  - 6 points are below the line of best fit.
  - The line of best fit passes through the following approximate coordinates:
    - (2 comma 2.9)
    - (8 comma 7.7)

The scatterplot shows the relationship between two variables, *x* and *y*. A line of best fit is also shown. For how many of the 11 data points does the line of best fit predict a greater *y*-value than the actual *y*-value?

Student-produced response: _____

A group of 10 botanists recorded data on the germination rates of a certain type of seed for one growing season. The scatterplot shows the relationship between the number of seeds planted, x, and the number of seeds that germinated, y, for each of the botanists. A line of best fit is also shown.



- The scatterplot has 10 data points.
- The data points are in a linear pattern trending up from left to right.
- A line of best fit is shown:
    - The line of best fit slants up from left to right.
    - The line of best fit passes through the following approximate coordinates:
        - (0 comma 0)
        - (200 comma 120)
        - (500 comma 300)

Which of the following is the best interpretation of the slope of the line of best fit in this context?

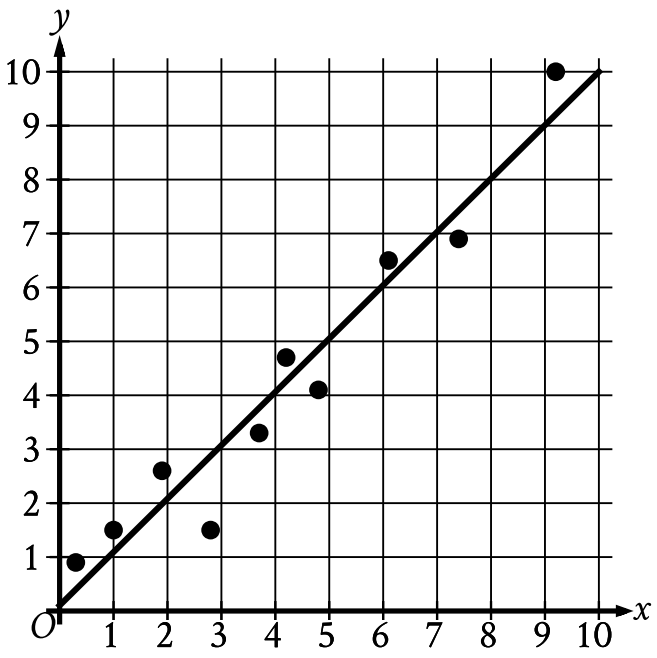
A. The number of seeds planted is predicted to increase by 60 seeds every 100 days.

B. The number of seeds planted is predicted to increase by 300 seeds every 100 days.

C. The number of seeds that germinate is predicted to increase by 60 seeds for every additional 100 seeds that are planted.

D. The number of seeds that germinate is predicted to increase by 300 seeds for every additional 100 seeds that are planted.

The scatterplot shows the relationship between two variables, x and y. A line of best fit for the data is also shown.



- The scatterplot has 10 data points.
- The data points are in a linear pattern trending up from left to right.
- A line of best fit is shown:
    - The line of best fit slants up from left to right.
    - 6 points are above the line of best fit.
    - 4 points are below the line of best fit.
    - The line of best fit passes through the following approximate coordinates:
        - (0 comma 0.1)
        - (5 comma 5.1)

For how many of the 10 data points is the actual y-value greater than the y-value predicted by the line of best fit?

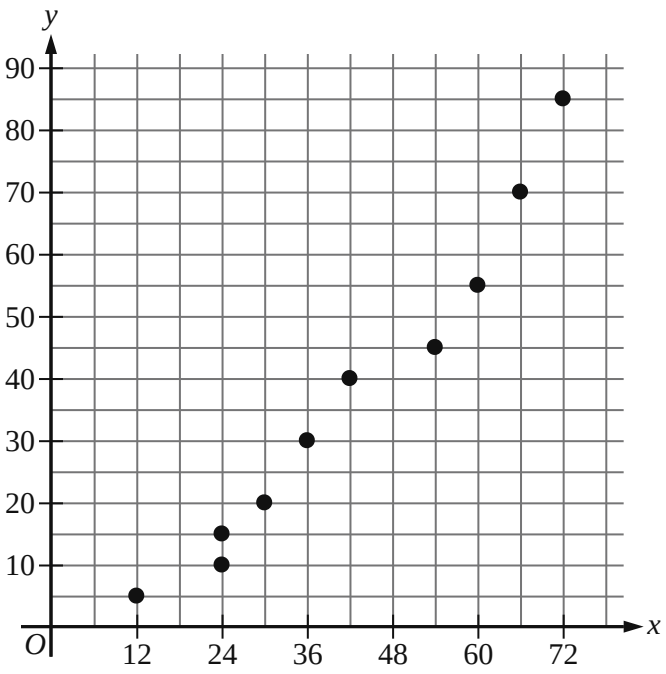
A. 3

B. 4

C. 6

D. 7

The owner of a restaurant supply store selects 10 cutting boards to put on sale. The graph shows the width x, in centimeters, of each cutting board and its corresponding sale price y, in dollars, where each value of y is an integer.



- The scatterplot has 10 data points.
- The data points are in a linear pattern trending up from left to right.
- The data points have the following coordinates:
  - (12 comma 5)
  - (24 comma 10)
  - (24 comma 15)
  - (30 comma 20)
  - (36 comma 30)
  - (42 comma 40)
  - (54 comma 45)
  - (60 comma 55)
  - (66 comma 70)
  - (72 comma 85)

According to the graph, which statement is true?

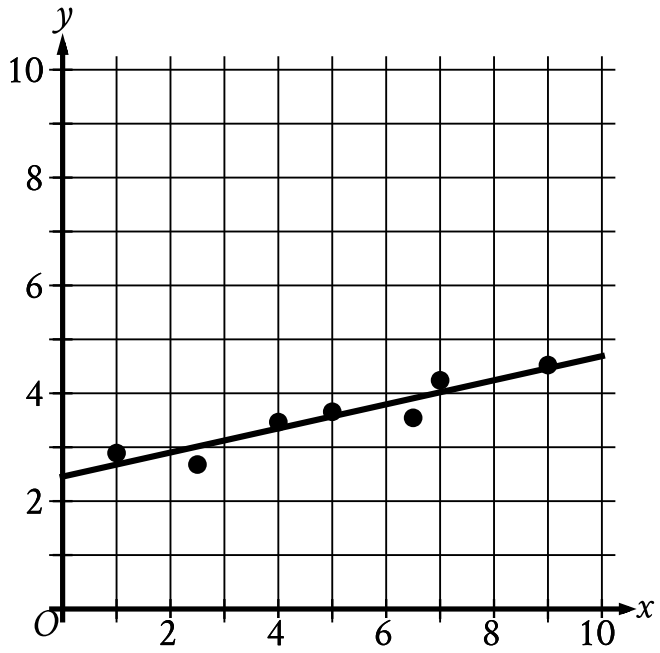
A. Cutting boards that have a width greater than 48 centimeters have an average sale price of $\$ 59.00$.

B. Cutting boards that have a width greater than 48 centimeters have an average sale price of $\$ 40.00$.

C. Cutting boards that have a width between 18 and 48 centimeters have an average sale price of $\$ 31.20$.

D. Cutting boards that have a width between 18 and 48 centimeters have an average sale price of $\$ 23.00$.

In the given scatterplot, a line of best fit for the data is shown.



- The scatterplot has 7 data points.
- The data points are in a linear pattern trending up from left to right.
- A line of best fit is shown:
  - The line of best fit slants up from left to right.
  - The line of best fit passes through the following approximate coordinates:
    - (0 comma 2.5)
    - (5 comma 3.6)
    - (10 comma 4.7)

Which of the following is closest to the slope of the line of best fit?

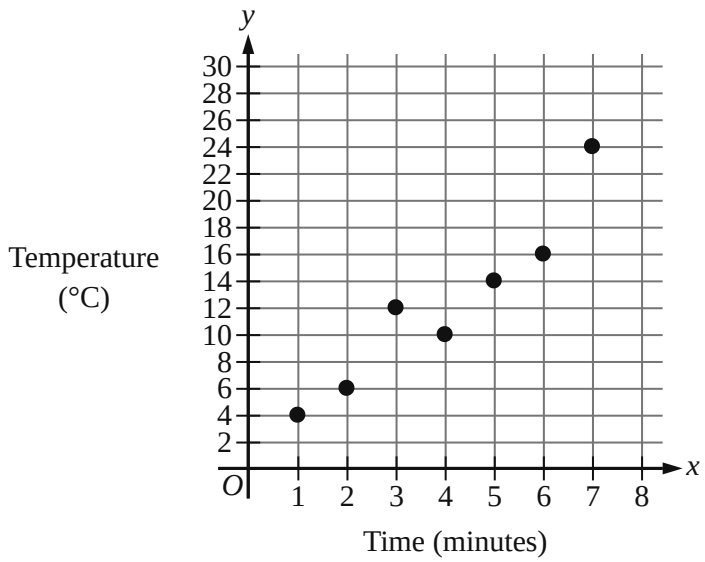
A. -2.45

B. -0.22

C. 0.22

D. 2.45

During a study, the temperature, in degrees Celsius °C, of the air in a chamber was recorded to the nearest integer at certain times. The scatterplot shows the recorded temperature y, in °C, of the air in the chamber x minutes after the start of the study.

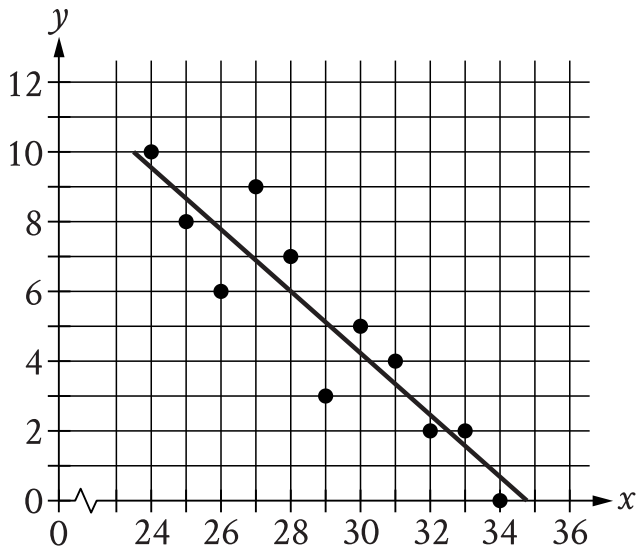


- The scatterplot has 7 data points.
- The data points are in a linear pattern trending up from left to right.
- The data points have the following coordinates:
  - (1 comma 4)
  - (2 comma 6)
  - (3 comma 12)
  - (4 comma 10)
  - (5 comma 14)
  - (6 comma 16)
  - (7 comma 24)

What was the average rate of change, in °C per minute, of the recorded temperature of the air in the chamber from $x = 5$ to $x = 7$?

Student-produced response: _____

The scatterplot shows the relationship between two variables, x and y. A line of best fit for the data is also shown.



- The scatterplot has 11 data points.
- The data points are in a linear pattern trending down from left to right.
- A line of best fit is shown:
  - The line of best fit slants down from left to right.
  - 6 points are above the line of best fit.
  - 5 points are below the line of best fit.
  - The line of best fit goes through the following approximate coordinates:
    - (28 comma 6)
    - (33 comma 1.5)

At $x = 25.5$, which of the following is closest to the y-value predicted by the line of best fit?

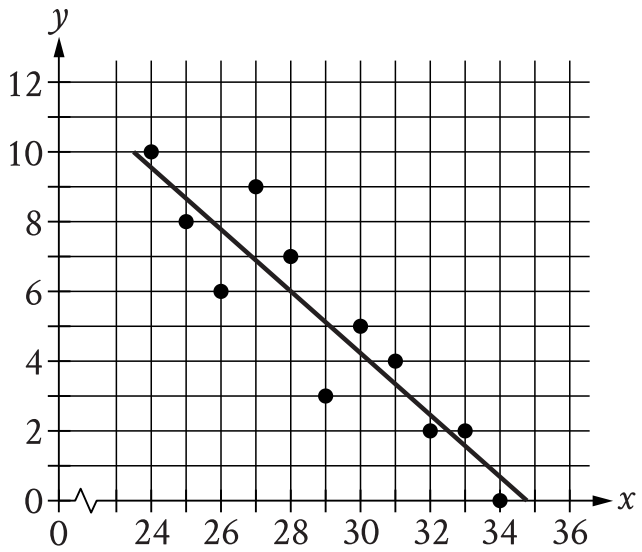
A. 6.2

B. 7.3

C. 8.2

D. 9.1

The scatterplot shows the relationship between two variables, x and y. A line of best fit for the data is also shown.



- The scatterplot has 11 data points.
- The data points are in a linear pattern trending down from left to right.
- A line of best fit is shown:
  - The line of best fit slants down from left to right.
  - 6 points are above the line of best fit.
  - 5 points are below the line of best fit.
  - The line of best fit goes through the following approximate coordinates:
    - (28 comma 6)
    - (33 comma 1.5)

At $x = 32$, which of the following is closest to the y-value predicted by the line of best fit?

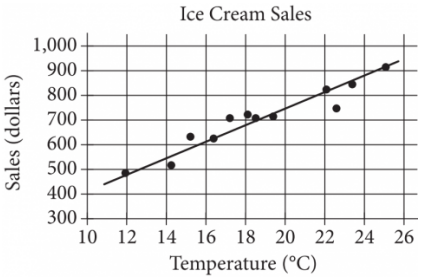
A. 0.4

B. 1.5

C. 2.4

D. 3.3

61 019531-DC · Hard



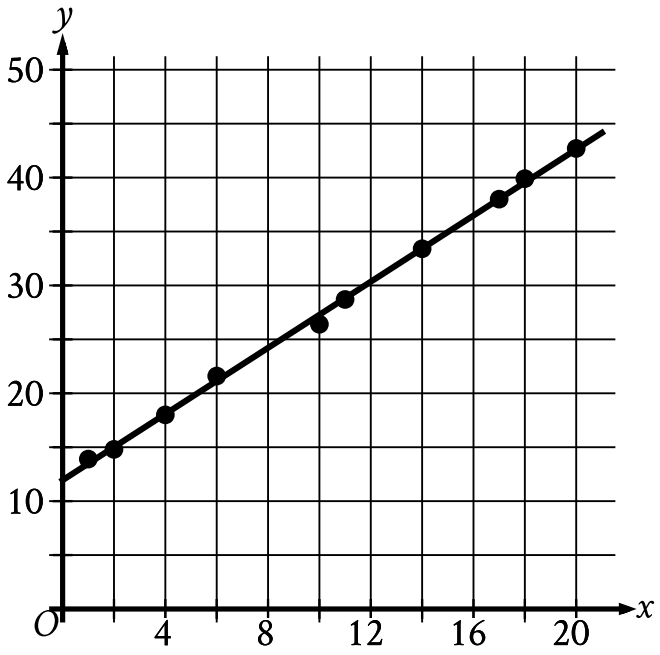
The scatterplot above shows a company's ice cream sales d, in dollars, and the high temperature t, in degrees Celsius (°C), on 12 different days. A line of best fit for the data is also shown. Which of the following could be an equation of the line of best fit?

A. $d = 0.03t + 402$

B. $d = 10t + 402$

C. $d = 33t + 300$

D. $d = 33t + 84$



- The scatterplot has 10 data points.
- The data points are in a linear pattern trending up from left to right.
- A line of best fit is shown:
    - The line of best fit slants up from left to right.
    - The line of best fit passes through the following approximate coordinates:
        - (0 comma 11.9)
        - (12 comma 30.3)
        - (20 comma 42.6)

The scatterplot shows the relationship between two variables, x and y, for data set E. A line of best fit is shown. Data set F is created by multiplying the y-coordinate of each data point from data set E by 3.9. Which of the following could be an equation of a line of best fit for data set F?

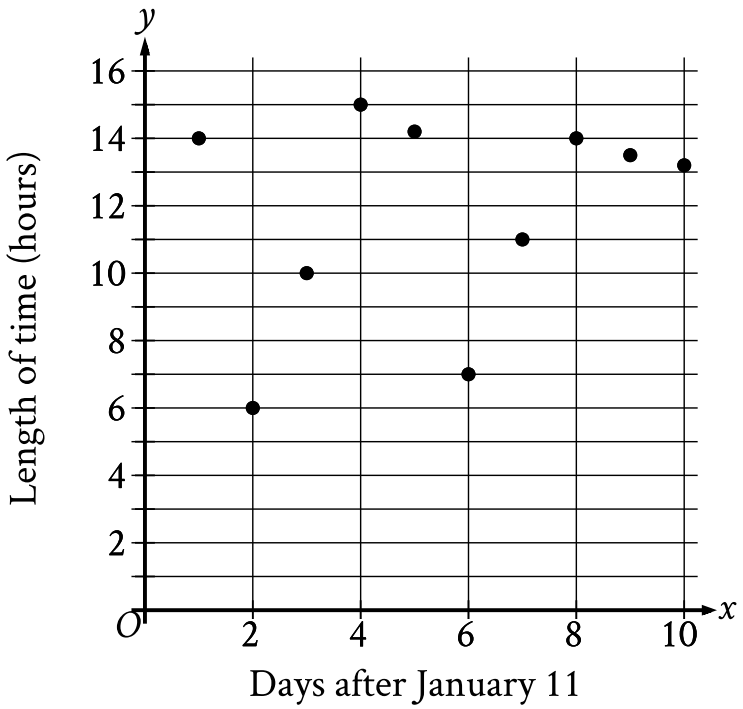
A. $y = 46.8 + 5.9x$

B. $y = 46.8 + 1.5x$

C. $y = 12 + 5.9x$

D. $y = 12 + 1.5x$

The scatterplot shows the relationship between the length of time y, in hours, a certain bird spent in flight and the number of days after January 11, x.



- The scatterplot has 10 data points.
- The data points are spread out.
- The data points have the following coordinates:
  - (1 comma 14)
  - (2 comma 6)
  - (3 comma 10)
  - (4 comma 15)
  - (5 comma 14.2)
  - (6 comma 7)
  - (7 comma 11)
  - (8 comma 14)
  - (9 comma 13.5)
  - (10 comma 13.2)

What is the average rate of change, in hours per day, of the length of time the bird spent in flight on January 13 to the length of time the bird spent in flight on January 15?

Student-produced response: _____

---

64 87f878d4-305d-414f-8da8-f430f23c0bfd · Hard

For $x > 0$, the function f is defined as follows:

$f(x)$ equals 201% of x

Which of the following could describe this function?

A. Decreasing exponential

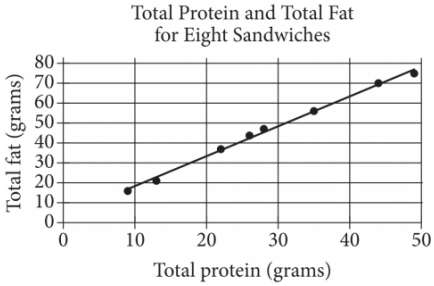
B. Decreasing linear

C. Increasing exponential

D. Increasing linear

---

65 08280-DC · Hard



The scatterplot above shows the numbers of grams of both total protein and total fat for eight sandwiches on a restaurant menu. The line of best fit for the data is also shown. According to the line of best fit, which of the following is closest to the predicted increase in total fat, in grams, for every increase of 1 gram in total protein?

A. 2.5

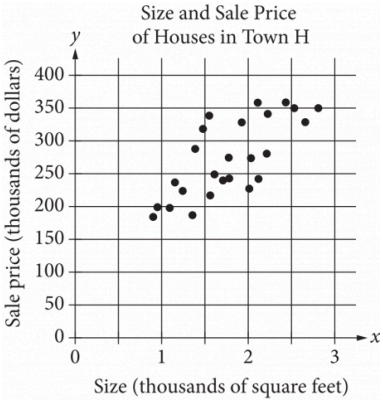
B. 2.0

C. 1.5

D. 1.0

---

66 14941-DC · Hard



The scatterplot above shows the size x and the sale price y of 25 houses for sale in Town H. Which of the following could be an equation for a line of best fit for the data?

A. $y = 200x + 100$

B. $y = 100x + 100$

C. $y = 50x + 100$

D. $y = 100x$

---

67 11107-DC · Hard

	Amount invested	Balance increase
Account A	$500	6% annual interest
Account B	$1,000	$25 per year

Two investments were made as shown in the table above. The interest in Account A is compounded once per year. Which of the following is true about the investments?

A. Account A always earns more money per year than Account B.

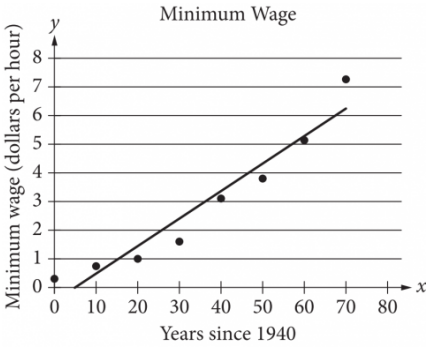
B. Account A always earns less money per year than Account B.

C. Account A earns more money per year than Account B at first but eventually earns less money per year.

D. Account A earns less money per year than Account B at first but eventually earns more money per year.

---

68 020386-DC · Hard



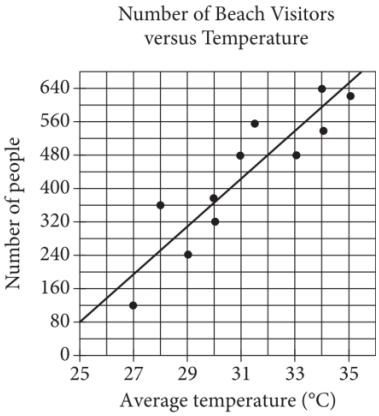
The scatterplot above shows the federal-mandated minimum wage every 10 years between 1940 and 2010. A line of best fit is shown, and its equation is $y = 0.096x - 0.488$. What does the line of best fit predict about the increase in the minimum wage over the 70-year period?

A. Each year between 1940 and 2010, the average increase in minimum wage was 0.096 dollars.

B. Each year between 1940 and 2010, the average increase in minimum wage was 0.49 dollars.

C. Every 10 years between 1940 and 2010, the average increase in minimum wage was 0.096 dollars.

D. Every 10 years between 1940 and 2010, the average increase in minimum wage was 0.488 dollars.

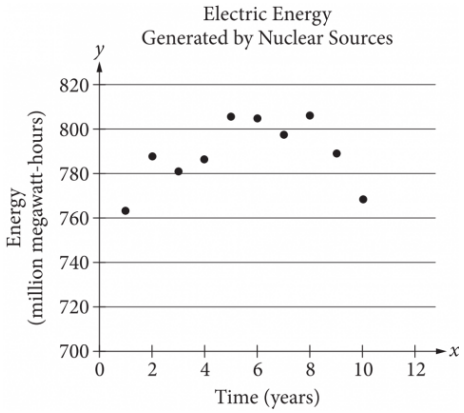


Each dot in the scatterplot above represents the temperature and the number of people who visited a beach in Lagos, Nigeria, on one of eleven different days. The line of best fit for the data is also shown. The line of best fit for the data has a slope of approximately 57. According to this estimate, how many additional people per day are predicted to visit the beach for each 5°C increase in average temperature?

Student-produced response: _____

70 10145-DC · Hard

The scatterplot below shows the amount of electric energy generated, in millions of megawatt-hours, by nuclear sources over a 10-year period.



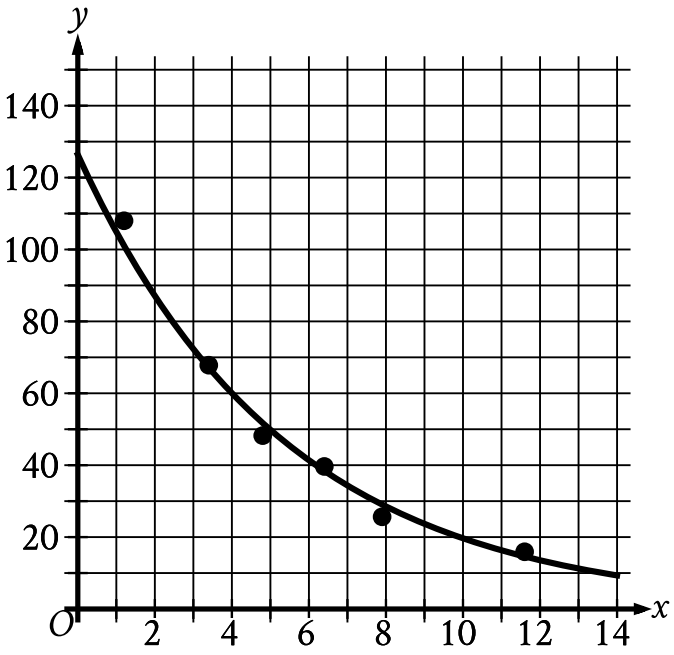
Of the following equations, which best models the data in the scatterplot?

A. $y = 1.674x^2 + 19.76x - 745.73$

B. $y = -1.674x^2 - 19.76x - 745.73$

C. $y = 1.674x^2 + 19.76x + 745.73$

D. $y = -1.674x^2 + 19.76x + 745.73$



- The scatterplot has 6 data points.
- The data points are in an exponential pattern trending down from left to right.
- A curve of best fit is shown:
    - The curve of best fit trends down from left to right.
    - The curve of best fit passes through the following approximate coordinates:
        - (0 comma 126)
        - (1 comma 105)
        - (5 comma 50)
        - (10 comma 20)

The scatterplot shows the relationship between two variables, x and y. An equation for the exponential model shown can be written as $y = ab^x$, where a and b are positive constants. Which of the following is closest to the value of b?

A. 0.83

B. 1.83

C. 18.36

D. 126.35